Top Quark Signal in the Cosmological Higgs Collider

1 Top quark propagators and Yukawa vertices

We calculate the Higgs four-point function generated by a top-quark box in exact de Sitter space. The top mass is constant, the chemical potential vanishes, and the external Higgs is a light spectator.

1.1 Higgs field and Majorana decomposition

The metric and scale factor are

$$\mathrm{d}s^2 = a^2(\tau)(-\mathrm{d}\tau^2 + \mathrm{d}\mathbf{x}^2), \qquad a(\tau) = -\frac{1}{H\tau}, \qquad \tau < 0. \tag{1}$$

We take $\sigma^a = (\mathbf{1}_2, \boldsymbol{\sigma})$ and $\bar{\sigma}^a = (\mathbf{1}_2, -\boldsymbol{\sigma})$. For $\mathbf{H} = (0, (h_0 + \delta h)/\sqrt{2})^T$, define

$$g_t = \frac{y_t}{\sqrt{2}}, \qquad m_t = g_t h_0, \qquad \tilde{\nu} = \frac{m_t}{H}, \qquad N_c = 3. \tag{2}$$

The notation follows the SMS manuscript: $\tilde{\nu}$ is the mass index, $\mathsf{a}, \mathsf{b} = \pm 1$ are SK labels, and $\eta = \pm 1$ labels the nonlocal branches. For the fermion, $\tilde{\nu} = m_t/H$; the vector relation $\tilde{\nu} = \sqrt{m_V^2/H^2 - 1/4}$ does not apply here. The massless Higgs bulk-to-boundary propagator is

$$G_{\mathsf{a}}(k;\tau) = \frac{H^2}{2k^3}(1 - \mathrm{i}\mathsf{a}k\tau)e^{\mathrm{i}\mathsf{a}k\tau}, \qquad \mathsf{a} = \pm 1. \tag{3}$$

Here m_H denotes the effective Higgs mass. This prescription assumes $|\dot{h}_0| \ll H|h_0|$ and $|m_H^2| \ll H^2$ over the relevant times [5]. The Higgs-to-curvature conversion is not included.

Spinor indices. The spinor indices are raised and lowered by

$$\begin{aligned} \epsilon_{12} = -\epsilon_{21} = 1, \qquad \epsilon^{12} = -\epsilon^{21} = -1, \\ \psi^\alpha = \epsilon^{\alpha\beta}\psi_\beta, \qquad \psi_\alpha = \epsilon_{\alpha\beta}\psi^\beta, \qquad \epsilon^{\alpha\beta}\epsilon_{\beta\gamma} = \delta^\alpha{}_\gamma. \end{aligned} \tag{4}$$

$$\psi\chi \equiv \psi^\alpha\chi_\alpha = -\epsilon^{\alpha\beta}\psi_\alpha\chi_\beta = \chi\psi, \qquad (\psi\psi)^\dagger = \psi^\dagger_{\dot{\alpha}}\psi^{\dagger\dot{\alpha}} = \epsilon^{\dot{\alpha}\dot{\beta}}\psi^\dagger_{\dot{\alpha}}\psi^\dagger_{\dot{\beta}}. \tag{5}$$

The symmetry $\psi\chi = \chi\psi$ uses both the Grassmann sign and the antisymmetry of epsilon; Hermitian conjugation reverses operator order.

Let t, t^c be the rescaled left-handed top and the left-handed charge conjugate of the right-handed top, respectively. The latter carries the conjugate gauge representation. The rescaled Dirac-field action is

$$\mathcal{L} = \mathrm{i}t^\dagger \bar{\sigma}^\mu \partial_\mu t + \mathrm{i}t^{c\dagger} \bar{\sigma}^\mu \partial_\mu t^c - a(m_t + g_t \delta h)(tt^c + \mathrm{h.c.}). \tag{6}$$

Our field redefinition is

$$t = \frac{\widetilde{\psi}_1 + \mathrm{i}\widetilde{\psi}_2}{\sqrt{2}}, \qquad t^c = \frac{\widetilde{\psi}_1 - \mathrm{i}\widetilde{\psi}_2}{\sqrt{2}}. \tag{7}$$

The bilinear $tt^c = t^\alpha (t^c)_\alpha$ is a Lorentz scalar. Its mixed terms cancel under this substitution, giving the coordinate Lagrangian density

$$\boxed{\mathcal{L} = \sum_{I=1}^{2} \left[\mathrm{i}\widetilde{\psi}_I^\dagger \bar{\sigma}^\mu \partial_\mu \widetilde{\psi}_I - \frac{a}{2}(m_t + g_t \delta h)(\widetilde{\psi}_I \widetilde{\psi}_I + \mathrm{h.c.}) \right].} \tag{8}$$

Gauge interactions are omitted, so the two copies are independent for each color. We calculate one copy and restore $2N_c$ in the box. The factor a in the mass and Yukawa terms is $a^4 (a^{-3/2})^2$.

We use $\epsilon_{12} = 1$, $\epsilon^{12} = -1$ and $\psi^\alpha = \epsilon^{\alpha\beta}\psi_\beta$, with the same rules for dotted indices. In numerical matrix products,

$$\boldsymbol{\epsilon} = (\epsilon_{\alpha\beta}) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \mathrm{i}\sigma^2, \qquad (\epsilon^{\alpha\beta}) = -\boldsymbol{\epsilon}. \tag{9}$$

Every field is Fourier transformed with $e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}}$ and measure $\mathrm{d}^3\mathbf{k}/(2\pi)^3$. A prime removes the overall momentum delta function.

1.2 Mode functions

For one Majorana copy the helicity spinors obey

$$(\boldsymbol{\sigma}\cdot\widehat{\mathbf{k}})h^s = s h^s, \qquad P_s = h^s h^{s\dagger} = \frac{\mathbf{1}_2 + s\boldsymbol{\sigma}\cdot\widehat{\mathbf{k}}}{2}. \tag{10}$$

With one copy and color suppressed, the operator expansion is

$$\widetilde{\psi}_\alpha(\tau, \mathbf{x}) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \sum_{s=\pm} \left[h_\alpha^s u_s b_{\mathbf{k}}^s e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}} + h_\gamma^{s\dagger} \bar{\sigma}^{0\dot{\gamma}\beta} \epsilon_{\alpha\beta} v_s^* b_{\mathbf{k}}^{s\dagger} e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{x}} \right]. \tag{11}$$

Here $\{b_{\mathbf{k}}^s, b_{\mathbf{k}'}^{s'\dagger}\} = (2\pi)^3 \delta^{ss'} \delta^3(\mathbf{k} - \mathbf{k}')$. The wave equations are

$$\mathrm{i}\partial_\tau u_s + sku_s = am_t v_s, \qquad \mathrm{i}\partial_\tau v_s - skv_s = am_t u_s, \tag{12}$$

$$u_s'' - aHu_s' + (k^2 + a^2 m_t^2 + \mathrm{i}saHk)u_s = 0, \qquad v_s'' - aHv_s' + (k^2 + a^2 m_t^2 - \mathrm{i}saHk)v_s = 0. \tag{13}$$

With $x = -k\tau$, the normalized Bunch–Davies modes are

$$\boxed{u_+ = v_- = \frac{\tilde{\nu}}{\sqrt{2x}} W_{-1/2,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}x), \qquad u_- = v_+ = \frac{\mathrm{i}}{\sqrt{2x}} W_{1/2,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}x).} \tag{14}$$

Here u_s, v_s are the two mode functions in the operator expansion (11); the creation term contains v_s^*. Their normalization is $|u_s|^2 + |v_s|^2 = 1$. In particular, $u_+ = v_-$ and $u_- = v_+$, with the relative i displayed above.

We use these Whittaker modes throughout. The time integrals will be reduced directly to the scalar chemical-potential seed of Ref. [3]; no additional mode decomposition is needed.

From the next subsection onward, we suppress the tildes: ψ denotes the conformally rescaled field $\widetilde{\psi}$, and D denotes its propagators. The original covariant spinor is no longer used. Thus the propagators below contain no external factors of $a^{-3/2}$; the Yukawa vertices retain the factor $a(\tau)$ in Eq. (8).

1.3 Propagators and vertices

We follow the three propagator types and SK ordering of You et al. [2] (v2), Eqs. (2.30), (2.33), and (2.36), specialized to $\widetilde{\lambda} = 0$ and $\widetilde{M}_R = \widetilde{\nu} = m_t/H$. All spinor indices below are lowered using Eq. (4). There is no overall factor of i in the propagator definitions.

Three types of propagators. With $1 = (\tau_1, \mathbf{k})$ and $2 = (\tau_2, -\mathbf{k})$, define

$$\begin{aligned} D_{\mathsf{ab}\,\alpha\dot{\beta}}(\mathbf{k};\tau_1,\tau_2) &= \langle T_C \psi_{\mathsf{a},\alpha}(1)\psi^\dagger_{\mathsf{b},\dot{\beta}}(2)\rangle', \\ \widehat{D}_{\mathsf{ab}\,\alpha\beta}(\mathbf{k};\tau_1,\tau_2) &= \langle T_C \psi_{\mathsf{a},\alpha}(1)\psi_{\mathsf{b},\beta}(2)\rangle', \\ \check{D}_{\mathsf{ab}\,\dot{\alpha}\dot{\beta}}(\mathbf{k};\tau_1,\tau_2) &= \langle T_C \psi^\dagger_{\mathsf{a},\dot{\alpha}}(1)\psi^\dagger_{\mathsf{b},\dot{\beta}}(2)\rangle'. \end{aligned} \tag{15}$$

Here T_C orders fermions on the closed time contour. The three $-+$ components are the unordered contractions $\langle\psi_1\psi_2^\dagger\rangle'$, $\langle\psi_1\psi_2\rangle'$, and $\langle\psi_1^\dagger\psi_2^\dagger\rangle'$, respectively. Using $u_{sj} = u_s(\tau_j, k)$, $v_{sj} = v_s(\tau_j, k)$ and $P_s = (\mathbf{1}_2 + s\sigma_k)/2$, $\sigma_k = \boldsymbol{\sigma}\cdot\hat{\mathbf{k}}$, their explicit helicity sums are

$$\begin{aligned} D_{-+} &= \sum_{s=\pm1} u_{s1}u^*_{s2}P_s, \\ \widehat{D}_{-+} &= -\sum_{s=\pm1} u_{s1}v^*_{s2}P_s\boldsymbol{\epsilon}, \\ \check{D}_{-+} &= \sum_{s=\pm1} v_{s1}u^*_{s2}\boldsymbol{\epsilon}P_s. \end{aligned} \tag{16}$$

These matrix expressions follow directly from Eq. (11): the creation wavefunction is $v_s^*\boldsymbol{\epsilon}h^{s*}$, and $h^s(\boldsymbol{\epsilon}h^{s*})^T = -P_s\boldsymbol{\epsilon}$. This fixes the epsilon signs when translating the helicity-spinor expressions of Ref. [2] to our lower-index convention.

Scalar propagator with chemical potential. Set $X = -k\tau_1$, $Y = -k\tau_2$. We use the scalar-valued chemical-potential propagator entering the Whittaker seed of Ref. [3], Eq. (96):

$$\begin{aligned} D^{\mathrm{sc},\kappa}_{-+}(k;\tau_1,\tau_2) &= \frac{e^{\mathrm{i}\pi\kappa}}{2k} W_{\kappa,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}X)W_{-\kappa,\mathrm{i}\tilde{\nu}}(2\mathrm{i}Y), \\ D^{\mathrm{sc},\kappa}_{+-}(k;\tau_1,\tau_2) &= \frac{e^{\mathrm{i}\pi\kappa}}{2k} W_{-\kappa,\mathrm{i}\tilde{\nu}}(2\mathrm{i}X)W_{\kappa,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}Y). \end{aligned} \tag{17}$$

The mass index $\tilde{\nu}$ is the $\tilde{\nu}$ of that reference, and $\kappa = \mathrm{i}h\tilde{\mu}$. We analytically continue to $\kappa = \pm 1/2$, or $h\tilde{\mu} = \mp\mathrm{i}/2$; the physical top chemical potential remains zero. *Scalar* here means the scalar

kernel of the chemical-potential seed, with its $1/(2k)$ normalization, not the covariant spin-zero de Sitter propagator $D^{(\mathrm{s})}$ used in the SMS manuscript. We retain the superscript sc,κ to distinguish these normalizations. Both Wightman expressions are continued as analytic functions; one must not conjugate the continued chemical potential anew. Write $D^{\mathrm{sc},+} = D^{\mathrm{sc},1/2}$ and $D^{\mathrm{sc},-} = D^{\mathrm{sc},-1/2}$. These superscripts are Whittaker indices, distinct from the SK subscripts a,b and the soft-branch label η.

The Whittaker recurrence relations give

$$\begin{aligned} A_X &= X\partial_X - \mathrm{i}X - \tfrac{1}{2}, & R_X &= -X\partial_X - \mathrm{i}X + \tfrac{1}{2}, \\ A_X W_{1/2,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}X) &= \tilde{\nu}^2 W_{-1/2,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}X), & R_X W_{-1/2,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}X) &= W_{1/2,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}X). \end{aligned} \tag{18}$$

Hence all three spinor propagators, for every SK pair, are

$$D_{\mathsf{ab}} = \frac{\mathrm{i}k}{\sqrt{XY}}\left[(R_X D^{\mathrm{sc},-}_{\mathsf{ab}})P_- - (A_X D^{\mathrm{sc},+}_{\mathsf{ab}})P_+\right], \tag{19}$$

$$\widehat{D}_{\mathsf{ab}} = -\frac{\tilde{\nu}k}{\sqrt{XY}}\left[D^{\mathrm{sc},-}_{\mathsf{ab}}P_+ + D^{\mathrm{sc},+}_{\mathsf{ab}}P_-\right]\boldsymbol{\epsilon}, \tag{20}$$

$$\check{D}_{\mathsf{ab}} = \frac{\tilde{\nu}k}{\sqrt{XY}}\boldsymbol{\epsilon}\left[D^{\mathrm{sc},+}_{\mathsf{ab}}P_+ + D^{\mathrm{sc},-}_{\mathsf{ab}}P_-\right]. \tag{21}$$

The operators act only on the scalar propagator to their right. The factors i and the signs include $e^{-\mathrm{i}\pi\kappa}$ from Eq. (17). The time derivatives implement Whittaker index shifts; they are not new derivative interactions.

The four SK components. For $\mathcal{D} = D, \widehat{D}, \check{D}$,

$$\mathcal{D}_{++} = \theta_{12}\mathcal{D}_{-+} + \theta_{21}\mathcal{D}_{+-}, \qquad \mathcal{D}_{--} = \theta_{21}\mathcal{D}_{-+} + \theta_{12}\mathcal{D}_{+-}, \tag{22}$$

where $\theta_{ij} = \theta(\tau_i - \tau_j)$. The scalar $D^{\mathrm{sc},\kappa}$ obeys the same ordering formula. Its Wightman functions coincide at equal times, so the first derivatives in Eq. (19) generate no delta function from differentiating the step functions. Fermionic signs are already contained in Eqs. (19)–(21); in particular the remaining field-type block is

$$D^{\bar{\mathrm{L}}\mathrm{L}}_{\mathsf{ab}}(\mathbf{k};\tau_1,\tau_2) = -D^{T}_{\mathsf{ba}}(-\mathbf{k};\tau_2,\tau_1). \tag{23}$$

Together with $D^{\mathrm{L}\bar{\mathrm{L}}} = D$, $D^{\mathrm{LL}} = \widehat{D}$ and $D^{\bar{\mathrm{L}}\bar{\mathrm{L}}} = \check{D}$, this specifies every propagator in the vertex-type sum.

Late-time expansion. At $x = -k\tau \ll 1$, define

$$c_\eta = \frac{2^{-1-2\mathrm{i}\eta\tilde{\nu}}}{\pi}\Gamma(\tfrac{1}{2} - \mathrm{i}\eta\tilde{\nu})^2, \qquad c \equiv c_+, \qquad c_- = c^*, \qquad |c| = \frac{1}{2\cosh\pi\tilde{\nu}}. \tag{24}$$

Expand the scalar propagators using

$$W_{\kappa,\mathrm{i}\tilde{\nu}}(z) = \sum_{\eta=\pm1}\frac{\Gamma(-2\mathrm{i}\eta\tilde{\nu})}{\Gamma(\frac{1}{2} - \mathrm{i}\eta\tilde{\nu} - \kappa)}z^{1/2+\mathrm{i}\eta\tilde{\nu}}\left[1 - \frac{\kappa z}{1+2\mathrm{i}\eta\tilde{\nu}} + O(z^2)\right]. \tag{25}$$

Substitute this in Eq. (17), apply A_X, R_X where required, and select the nonanalytic momentum powers. For a soft propagator momentum $\mathbf{k}_s$, with $k_s = |\mathbf{k}_s|$ and $|k_s\tau_1|, |k_s\tau_2| \ll 1$, the leading and subleading nonanalytic terms are

$$\begin{aligned}
&\Big[D^{\mathrm{L}\overline{\mathrm{L}}}_{\mathsf{ab}\,\alpha\dot\beta}(\mathbf{k}_s;\tau_1,\tau_2)\Big]_{\mathrm{NonL}} = \Big[D_{\mathsf{ab}\,\alpha\dot\beta}(\mathbf{k}_s;\tau_1,\tau_2)\Big]_{\mathrm{NonL}} \\
&= -\sum_{\eta=\pm1} \frac{2^{-2-2\mathrm{i}\eta\tilde\nu}\Gamma(\frac12-\mathrm{i}\eta\tilde\nu)^2}{\pi}(\tau_1\tau_2)^{\mathrm{i}\eta\tilde\nu}\left[k_s^{2\mathrm{i}\eta\tilde\nu}\boldsymbol{\sigma}\cdot\widehat{\mathbf{k}}_s + \frac{\mathrm{i}(\tau_1-\tau_2)}{1+2\mathrm{i}\eta\tilde\nu}k_s^{1+2\mathrm{i}\eta\tilde\nu}\mathbf{1}_2\right]_{\alpha\dot\beta} + \cdots, \qquad (26)\\
&\Big[\widehat{D}_{\mathsf{ab}\,\alpha\beta}(\mathbf{k}_s;\tau_1,\tau_2)\Big]_{\mathrm{NonL}} \\
&= -\sum_{\eta=\pm1} \frac{\eta\,2^{-2-2\mathrm{i}\eta\tilde\nu}\Gamma(\frac12-\mathrm{i}\eta\tilde\nu)^2}{\pi}(\tau_1\tau_2)^{\mathrm{i}\eta\tilde\nu}\left[k_s^{2\mathrm{i}\eta\tilde\nu}(\boldsymbol{\sigma}\cdot\widehat{\mathbf{k}}_s)\boldsymbol{\epsilon} + \frac{\mathrm{i}(\tau_1+\tau_2)}{1+2\mathrm{i}\eta\tilde\nu}k_s^{1+2\mathrm{i}\eta\tilde\nu}\boldsymbol{\epsilon}\right]_{\alpha\beta} + \cdots, \qquad (27)\\
&\Big[\check{D}_{\mathsf{ab}\,\dot\alpha\dot\beta}(\mathbf{k}_s;\tau_1,\tau_2)\Big]_{\mathrm{NonL}} \\
&= -\sum_{\eta=\pm1} \frac{\eta\,2^{-2-2\mathrm{i}\eta\tilde\nu}\Gamma(\frac12-\mathrm{i}\eta\tilde\nu)^2}{\pi}(\tau_1\tau_2)^{\mathrm{i}\eta\tilde\nu}\left[k_s^{2\mathrm{i}\eta\tilde\nu}\boldsymbol{\epsilon}(\boldsymbol{\sigma}\cdot\widehat{\mathbf{k}}_s) - \frac{\mathrm{i}(\tau_1+\tau_2)}{1+2\mathrm{i}\eta\tilde\nu}k_s^{1+2\mathrm{i}\eta\tilde\nu}\boldsymbol{\epsilon}\right]_{\dot\alpha\dot\beta} + \cdots. \qquad (28)
\end{aligned}$$

The two displayed terms scale as $k_s^{2\mathrm{i}\eta\tilde\nu}$ and $k_s^{1+2\mathrm{i}\eta\tilde\nu}$; the omitted terms start at $k_s^{2+2\mathrm{i}\eta\tilde\nu}$ at fixed endpoint times. The ellipses denote only these higher-order soft corrections. Both mass branches are already included in $\sum_{\eta=\pm1}$: no additional complex-conjugate term or $\tilde\nu \to -\tilde\nu$ term is implied by the ellipses. Here $\boldsymbol{\epsilon} = \mathrm{i}\sigma^2$, and the identity and Pauli matrices carry the displayed spinor indices. These nonlocal terms are the same for all four SK assignments ab, following the cutting prescription of Ref. [2], Eq. (3.13) and Appendix D.3.

Yukawa vertices. There are two Yukawa vertices, $\delta h\psi\psi$ and $\delta h\psi^\dagger\psi^\dagger$. Their spin tensors, labelled L and $\overline{\mathrm{L}}$ respectively, are

$$J_{\mathrm{L}}^{\alpha\beta} = -\epsilon^{\alpha\beta}, \qquad J_{\overline{\mathrm{L}}}^{\dot\alpha\dot\beta} = +\epsilon^{\dot\alpha\dot\beta}, \qquad J_{\mathrm{L}} = \boldsymbol{\epsilon}, \quad J_{\overline{\mathrm{L}}} = -\boldsymbol{\epsilon}. \qquad (29)$$

On SK branch $\mathsf{a} = \pm1$ their amputated rules are $-\mathrm{i}\mathsf{a}g_t a(\tau)J_r$. The two identical fermion slots cancel the $1/2$ in the interaction. The equivalent calculation with unamputated bilinears and explicit Wick pairings is given in Section 2.1.

2 The four-point box

2.1 Vertex types, Wick contractions and Majorana counting

The two unamputated interaction insertions are

$$\begin{aligned}
&-\frac{\mathrm{i}\mathsf{a}g_t a(\tau)}{2}\delta h\,(-\epsilon^{\alpha\beta})\psi_\alpha\psi_\beta, \\
&-\frac{\mathrm{i}\mathsf{a}g_t a(\tau)}{2}\delta h\,\epsilon^{\dot\alpha\dot\beta}\psi^\dagger_{\dot\alpha}\psi^\dagger_{\dot\beta}. \qquad (30)
\end{aligned}$$

We label the two field types by $r, s \in \{\mathrm{L}, \overline{\mathrm{L}}\}$, where L denotes ψ and $\overline{\mathrm{L}}$ denotes $\psi^\dagger$. Each vertex can be of either Yukawa type. For a fixed cycle the 16 choices are grouped as

$(\delta h\psi\psi, \delta h\psi^\dagger\psi^\dagger)$ vertices	$(4,0)$	$(3,1)$	$(2,2)$	$(1,3)$	$(0,4)$
number of type assignments	1	4	6	4	1

For a fixed choice of vertex types around $i \to j \to k \to l \to i$, write $J_i = J_{r_i}$ and $D_{ij} = D_{ij}^{r_i r_j}$ in this paragraph. Each matrix product contracts one spinor index:

$$(J_i D_{ij})^\alpha{}_\beta = \sum_{\gamma=1}^{2} (J_i)^{\alpha\gamma} (D_{ij})_{\gamma\beta}.$$

The indices are dotted or undotted according to the field at each endpoint. Multiplying the four segments and closing the last index with the first gives

$$\mathcal{T}_{ijkl} = \mathrm{tr}_2[(J_i D_{ij})(J_j D_{jk})(J_k D_{kl})(J_l D_{li})]\,. \tag{31}$$

Here tr_2 sums only the two spinor components. We only calculated the diagram '1234', and other diagram can be obtained by external permutation.

For one Majorana copy and one color, and for the full top loop, respectively,

$$\langle \delta h_{\mathbf{k}_1} \delta h_{\mathbf{k}_2} \delta h_{\mathbf{k}_3} \delta h_{\mathbf{k}_4} \rangle' = -\frac{g_t^4}{2} \sum_{(i,j,k,l=L,\bar{L})} \sum_{\mathsf{a}_1,\ldots,\mathsf{a}_4=\pm} \left(\prod_{v=1}^{4} \mathsf{a}_v \right) \int_{-\infty}^{0} \prod_{v=1}^{4} [\mathrm{d}\tau_v\, a(\tau_v) G_{\mathsf{a}_v}(k_v;\tau_v)] \int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \mathcal{T}_{ijkl} \tag{32}$$

The minus sign is the closed-fermion-loop sign.

2.2 The double-soft cut and the cutting rule

The double-soft factorization [4] selects four s-channel cycles as $\mathbf{k}_s = \mathbf{k}_1 + \mathbf{k}_2 \to 0$: In 1234, the soft momenta are $\mathbf{q}$ and $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$, while the right hard momentum is exactly $\mathbf{q} - \mathbf{k}_4 = \mathbf{q} + \mathbf{k}_s + \mathbf{k}_3$.

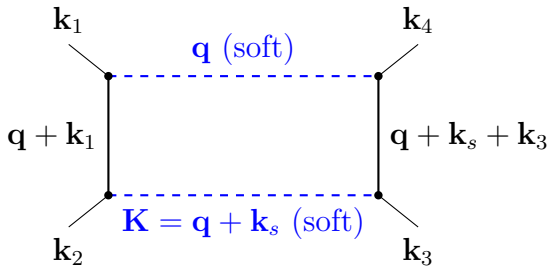


Figure 1: All momentum labels are vectors. Dashed lines select the double-soft region; solid internal lines retain complete hard time dependence.

In particular, the normal soft propagator is nonzero and retains σ_k. The hard lines keep their full time dependence and all SK components.

3 Oscillatory contribution

3.1 Unintegrated box and soft expansion

We start with the fixed cycle 1234, keeping both the loop and time integrals explicit. For each choice $r_1, r_2, r_3, r_4 \in \{\mathrm{L}, \overline{\mathrm{L}}\}$, its integrand is

$$\begin{aligned}\mathcal{G}^{\mathrm{top}}_{1234} =& \; -N_c g_t^4 \sum_{\mathsf{a}_1,\ldots,\mathsf{a}_4=\pm} \prod_{v=1}^{4} \mathsf{a}_v \int_{-\infty}^{0} \prod_{v=1}^{4} [\mathrm{d}\tau_v \, a(\tau_v) G_{\mathsf{a}_v}(k_v;\tau_v)] \int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \sum_{r_1,\ldots,r_4} \mathcal{T}_{1234}, \\ \mathcal{T}_{1234} =& \; \mathrm{tr}_2[D_{12} J_{r_2} D_{23} J_{r_3} D_{34} J_{r_4} D_{41} J_{r_1}], \qquad D_{ij} = D^{r_i r_j}_{\mathsf{a}_i \mathsf{a}_j}(\mathbf{p}_{ij};\tau_i,\tau_j). \end{aligned} \tag{33}$$

The minus sign is the closed-loop sign. Both Majorana copies and all colors have been included, so the single-copy coefficient $-g_t^4/2$ has become $-N_c g_t^4$. The sum over r_v changes the vertex types; it does not include external permutations.

Use the centered momenta

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{k}_L + \tfrac{1}{2}\mathbf{k}_s, & \mathbf{k}_2 &= -\mathbf{k}_L + \tfrac{1}{2}\mathbf{k}_s, \\ \mathbf{k}_3 &= \mathbf{k}_R - \tfrac{1}{2}\mathbf{k}_s, & \mathbf{k}_4 &= -\mathbf{k}_R - \tfrac{1}{2}\mathbf{k}_s, \\ \boldsymbol{\ell} &= \mathbf{q} + \tfrac{1}{2}\mathbf{k}_s, & \mathbf{K} &= \mathbf{q} + \mathbf{k}_s, \qquad \mathbf{n}_L = \widehat{\mathbf{k}}_L, \quad \mathbf{n}_R = \widehat{\mathbf{k}}_R, \quad \mathbf{n}_s = \widehat{\mathbf{k}}_s. \end{aligned} \tag{34}$$

Thus $\mathbf{p}_{12} = \mathbf{k}_L + \boldsymbol{\ell}$, $\mathbf{p}_{23} = \mathbf{K}$, $\mathbf{p}_{34} = \mathbf{k}_R + \boldsymbol{\ell}$, and $\mathbf{p}_{41} = \mathbf{q}$. We expand in $q/k_{L,R}$ and $k_s/k_{L,R}$, but retain the nonanalytic powers of q and K and their unit vectors. At the final accuracy, $k_L = k_{12}/2$, $k_R = k_{34}/2$, with $k_{ij} = k_i + k_j$.

For the $+\mathrm{i}\tilde{\nu}$ branch, the terms retained at the two soft endpoints can be factored as

$$D^{rs}_{\mathrm{NonL},+}(k;\tau,\tau') \simeq -\frac{c}{2}(k^2\tau\tau')^{\mathrm{i}\tilde{\nu}} L^+_r(k,\tau)\sigma_k U^+_s(k,\tau'). \tag{35}$$

Here $D^{rs}_{\mathrm{NonL},+}$ denotes the $\eta = +1$ nonlocal branch, $c = c_+$ is defined in Eq. (24), and $\sigma_k = \boldsymbol{\sigma}\cdot\widehat{\mathbf{k}}$. The endpoint matrices through first order are

$$\begin{aligned} L^+_{\mathrm{L}} &= \mathbf{1}_2 + \frac{\mathrm{i}\tau\boldsymbol{\sigma}\cdot\mathbf{k}}{1+2\mathrm{i}\tilde{\nu}}, & L^+_{\overline{\mathrm{L}}} &= \boldsymbol{\epsilon}\left(\mathbf{1}_2 - \frac{\mathrm{i}\tau\boldsymbol{\sigma}\cdot\mathbf{k}}{1+2\mathrm{i}\tilde{\nu}}\right), \\ U^+_{\mathrm{L}} &= \left(\mathbf{1}_2 + \frac{\mathrm{i}\tau\boldsymbol{\sigma}\cdot\mathbf{k}}{1+2\mathrm{i}\tilde{\nu}}\right)\boldsymbol{\epsilon}, & U^+_{\overline{\mathrm{L}}} &= \mathbf{1}_2 - \frac{\mathrm{i}\tau\boldsymbol{\sigma}\cdot\mathbf{k}}{1+2\mathrm{i}\tilde{\nu}}. \end{aligned} \tag{36}$$

The omitted terms at each endpoint are $O(k^2\tau^2)$. These products reproduce the $\eta = +1$ terms in Eqs. (26)–(28) through first order. The leading nonzero box signal also requires products of first-order corrections at opposite hard ends.

First fix the four vertex types $r_1, r_2, r_3, r_4 \in \{\mathrm{L}, \overline{\mathrm{L}}\}$ in the cycle 1234. The corresponding left and right hard spin matrices are

$$\begin{aligned} V_L^{r_1 r_2} &= -\frac{1}{2} U_{r_1}(q,\tau_1) J_{r_1} D^{r_1 r_2}_{\mathsf{a}_1\mathsf{a}_2}(\mathbf{k}_L + \boldsymbol{\ell};\tau_1,\tau_2) J_{r_2} L_{r_2}(K,\tau_2), \\ V_R^{r_3 r_4} &= -\frac{1}{2} U_{r_3}(K,\tau_3) J_{r_3} D^{r_3 r_4}_{\mathsf{a}_3\mathsf{a}_4}(\mathbf{k}_R + \boldsymbol{\ell};\tau_3,\tau_4) J_{r_4} L_{r_4}(q,\tau_4). \end{aligned} \tag{37}$$

Here $J_{\rm L} = \boldsymbol{\epsilon}$ and $J_{\overline{\rm L}} = -\boldsymbol{\epsilon}$ are the vertex matrices; $L_r = L_r^+$ and $U_r = U_r^+$ are soft endpoint matrices, not additional vertices. The factors $-1/2$ come from the soft lines. No vertex-type sum is implicit in these two matrices. Summing the 16 choices of the four vertex types gives the cut spin factor

$$c^2(qK)^{2i\tilde{\nu}} \prod_v (-\tau_v)^{i\tilde{\nu}} \sum_{r_1,r_2,r_3,r_4={\rm L},\overline{\rm L}} {\rm tr}_2[V_L^{r_1r_2}\sigma_K V_R^{r_3r_4}\sigma_q]. \tag{38}$$

To display the remaining time measure, define only the external polynomial and phase

$$E_{\mathsf{a}}(k,\tau) = (1 - i\mathsf{a}k\tau)e^{i\mathsf{a}k\tau}, \qquad G_{\mathsf{a}}(k;\tau) = \frac{H^2}{2k^3}E_{\mathsf{a}}(k,\tau).$$

Substitution in Eq. (33) gives

$$\mathcal{G}^{\rm top}_{1234}\Big|_{\rm clock} = -\frac{N_c g_t^4 H^4}{16\prod_i k_i^3}\Bigg[c^2 \sum_{\mathsf{a}_1,\ldots,\mathsf{a}_4}\prod_v \mathsf{a}_v \int_{-\infty}^0 \prod_v [{\rm d}\tau_v\,(-\tau_v)^{i\tilde{\nu}-1}E_{\mathsf{a}_v}(k_v,\tau_v)] \times \int \frac{{\rm d}^3\mathbf{q}}{(2\pi)^3}(qK)^{2i\tilde{\nu}} \sum_{r_1,\ldots,r_4} {\rm tr}_2[V_L^{r_1r_2}\sigma_K V_R^{r_3r_4}\sigma_q] + {\rm c.c.}\Bigg]. \tag{39}$$

No time integration is implicit in V_L, V_R. The complex conjugate supplies the opposite same-sign soft branch. We keep orders zero and one at each hard end. The first surviving full-box term will be the product of the two first-order terms.

Let $\mathbf{p} = \mathbf{k}_L + \boldsymbol{\ell}$ and $\sigma_p = \boldsymbol{\sigma}\cdot\hat{\mathbf{p}}$. Expanding the endpoint matrices in Eq. (36), the four terms in the left type sum are

$$\begin{aligned}
2V_L^{\rm LL} &= \left(\mathbf{1}_2 + \frac{i\tau_1\boldsymbol{\sigma}\cdot\mathbf{q}}{1+2i\tilde{\nu}}\right) D^{\rm LL}\boldsymbol{\epsilon}\left(\mathbf{1}_2 + \frac{i\tau_2\boldsymbol{\sigma}\cdot\mathbf{K}}{1+2i\tilde{\nu}}\right)\\
2V_L^{\rm L\overline{L}} &= \left(\mathbf{1}_2 + \frac{i\tau_1\boldsymbol{\sigma}\cdot\mathbf{q}}{1+2i\tilde{\nu}}\right) D^{\rm L\overline{L}}\left(\mathbf{1}_2 - \frac{i\tau_2\boldsymbol{\sigma}\cdot\mathbf{K}}{1+2i\tilde{\nu}}\right)\\
2V_L^{\overline{\rm L}{\rm L}} &= \left(\mathbf{1}_2 - \frac{i\tau_1\boldsymbol{\sigma}\cdot\mathbf{q}}{1+2i\tilde{\nu}}\right) \boldsymbol{\epsilon} D^{\overline{\rm L}{\rm L}}\boldsymbol{\epsilon}\left(\mathbf{1}_2 + \frac{i\tau_2\boldsymbol{\sigma}\cdot\mathbf{K}}{1+2i\tilde{\nu}}\right)\\
2V_L^{\overline{\rm L}\overline{\rm L}} &= \left(\mathbf{1}_2 - \frac{i\tau_1\boldsymbol{\sigma}\cdot\mathbf{q}}{1+2i\tilde{\nu}}\right) \boldsymbol{\epsilon} D^{\overline{\rm L}\overline{\rm L}}\left(\mathbf{1}_2 - \frac{i\tau_2\boldsymbol{\sigma}\cdot\mathbf{K}}{1+2i\tilde{\nu}}\right).
\end{aligned} \tag{40}$$

This equation is used through first order in the soft momenta; all four hard propagators carry the arguments in Eq. (37).

The four vertex types reduce the spin matrices to two scalar coefficients, expressed directly through Eq. (17). With $X = -p\tau_1$, $Y = -p\tau_2$, define

$$\begin{aligned}
\mathcal{K}_{0;\mathsf{ab}}(p;\tau_1,\tau_2) &= \frac{ip}{2\sqrt{XY}}\left[(A_X - i\tilde{\nu})D^{{\rm sc},+}_{\mathsf{ab}} + (R_X + i\tilde{\nu})D^{{\rm sc},-}_{\mathsf{ab}}\right],\\
\mathcal{K}_{1;\mathsf{ab}}(p;\tau_1,\tau_2) &= \frac{ip}{2\sqrt{XY}}\left[(R_X - i\tilde{\nu})D^{{\rm sc},-}_{\mathsf{ab}} - (A_X + i\tilde{\nu})D^{{\rm sc},+}_{\mathsf{ab}}\right].
\end{aligned} \tag{41}$$

Every scalar propagator has arguments $(p;\tau_1,\tau_2)$. These are hard spin-matrix coefficients, not new propagators. For the fixed 1234 ordering the type sum, before any integration, is

$$\begin{aligned}
\sum_{r_1,r_2} V_L^{r_1r_2} = &-\mathcal{K}_{0;\mathsf{a}_1\mathsf{a}_2}(p;\tau_1,\tau_2)\sigma_p\\
&+\frac{i}{1+2i\tilde{\nu}}\left[\tau_1\mathcal{K}_{1;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\boldsymbol{\sigma}\cdot\mathbf{q} + \tau_2\mathcal{K}_{1;\mathsf{a}_2\mathsf{a}_1}(k_L;\tau_2,\tau_1)\boldsymbol{\sigma}\cdot\mathbf{K}\right] + O({\rm soft}^2).
\end{aligned} \tag{42}$$

Here $\mathbf{p} = \mathbf{k}_L + \boldsymbol{\ell}$. In the first term both the magnitude and direction must be expanded:

$$\begin{aligned} -\mathcal{K}_0(p)\sigma_p = & -\mathcal{K}_0(k_L)\boldsymbol{\sigma}\cdot\mathbf{n}_L \\ & -\frac{1}{k_L}\boldsymbol{\sigma}\cdot\Big[\mathcal{K}_0(k_L)\boldsymbol{\ell} + (p\partial_p - 1)\mathcal{K}_0(p)\big|_{p=k_L}\,\mathbf{n}_L(\mathbf{n}_L\cdot\boldsymbol{\ell})\Big] + O(\text{soft}^2). \end{aligned} \tag{43}$$

The derivative is at fixed times. In particular,

$$\sum_{r_1,r_2} V_L^{r_1 r_2,(0)} = -\mathcal{K}_{0;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\boldsymbol{\sigma}\cdot\mathbf{n}_L \neq 0. \tag{44}$$

The right end follows from $(1,2,L,\mathbf{q},\mathbf{K}) \to (3,4,R,\mathbf{K},\mathbf{q})$. The hard propagators retain their complete SK time dependence.

3.2 Integrand cancellation and the spin trace

Keep each time and SK label attached to its physical external vertex. In particular, the exact Higgs factors form the common weight

$$\mathcal{E}_{\{\mathsf{a}\}}(\{\tau\};\{k\}) = \prod_{v=1}^{4}[\mathsf{a}_v(-\tau_v)^{\mathrm{i}\tilde{\nu}-1}E_{\mathsf{a}_v}(k_v,\tau_v)]. \tag{45}$$

This weight is independent of the loop momentum $\mathbf{q}$. It is kept outside the loop integral without expanding any E_{a_v} in k_s. The normalization $1/\prod_v k_v^3$ is also kept exact.

Four actual diagrams, with fixed external labels. The four s-cut orderings are

$$\mathcal{G}^{\text{top}}\Big|_{s,\text{clock}} = (\mathcal{G}^{\text{top}}_{1234} + \mathcal{G}^{\text{top}}_{1243} + \mathcal{G}^{\text{top}}_{1342} + \mathcal{G}^{\text{top}}_{1432})\Big|_{\text{clock}}. \tag{46}$$

Write $V_{12} = \sum_{r_1,r_2} V_L^{r_1 r_2}$ and $V_{34} = \sum_{r_3,r_4} V_R^{r_3 r_4}$ for the matrices already defined in Eq. (37). When the left connection is reversed, the q endpoint is at vertex 2 and the K endpoint at vertex 1. The reversed right connection similarly puts the K endpoint at vertex 4. Their matrices are

$$\begin{aligned} V_{21} &= -\frac{1}{2}\sum_{r_1,r_2} U_{r_2}(q,\tau_2)J_{r_2}D^{r_2 r_1}_{\mathsf{a}_2\mathsf{a}_1}(-\mathbf{k}_L+\boldsymbol{\ell};\tau_2,\tau_1)J_{r_1}L_{r_1}(K,\tau_1), \\ V_{43} &= -\frac{1}{2}\sum_{r_3,r_4} U_{r_4}(K,\tau_4)J_{r_4}D^{r_4 r_3}_{\mathsf{a}_4\mathsf{a}_3}(-\mathbf{k}_R+\boldsymbol{\ell};\tau_4,\tau_3)J_{r_3}L_{r_3}(q,\tau_3). \end{aligned} \tag{47}$$

These are different vertex connections, not averages of the integrand of 1234. For example, $E_{\mathsf{a}_1}(k_1,\tau_1)$ stays at vertex 1 in both V_{12} and V_{21}. The four diagrams correspond respectively to

$$(1234, 1243, 1342, 1432) \;\longleftrightarrow\; ((12,34),(12,43),(21,34),(21,43)). \tag{48}$$

Thus their common external factor multiplies four separate spin traces:

$$\begin{aligned} c^2(qK)^{2\mathrm{i}\tilde{\nu}}\mathcal{E}_{\{\mathsf{a}\}}\Big\{ & \mathrm{tr}_2[V_{12}\sigma_K V_{34}\sigma_q] + \mathrm{tr}_2[V_{12}\sigma_K V_{43}\sigma_q] \\ & + \mathrm{tr}_2[V_{21}\sigma_K V_{34}\sigma_q] + \mathrm{tr}_2[V_{21}\sigma_K V_{43}\sigma_q]\Big\}. \end{aligned} \tag{49}$$

Sum over the four SK labels and integrate this expression with $\prod_v \mathrm{d}\tau_v\, \mathrm{d}^3\mathbf{q}/(2\pi)^3$, add its complex conjugate, and multiply by $-N_c g_t^4 H^4/(16\prod_v k_v^3)$.

The scalar kernel has the reversal identity

$$\mathcal{K}_{0;\mathsf{ab}}(p;\tau_1,\tau_2) = \mathcal{K}_{0;\mathsf{ba}}(p;\tau_2,\tau_1). \tag{50}$$

It follows from fermionic propagator reversal and the vertex tensors; no exchange of integration variables is being performed here. Applying this identity to the two physical connections gives

$$\begin{aligned} V_{12}^{(0)} &= -\mathcal{K}_{0;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\boldsymbol{\sigma}\cdot\mathbf{n}_L, & V_{21}^{(0)} &= +\mathcal{K}_{0;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\boldsymbol{\sigma}\cdot\mathbf{n}_L, \\ V_{12}^{(0)} + V_{21}^{(0)} &= 0, & V_{34}^{(0)} + V_{43}^{(0)} &= 0. \end{aligned} \tag{51}$$

Because the exact external weight is common to all four traces, this cancellation is pointwise even before the time integrations. It removes every product with a zeroth-order end, including $(0,0)$, $(1,0)$, $(0,1)$, $(2,0)$ and $(0,2)$. In particular,

$$\boxed{\mathcal{G}^{\mathrm{box}}_{s,\mathrm{clock}}\Big|_{k_s^{3\pm 4\mathrm{i}\tilde{\nu}}} = 0.} \tag{52}$$

A single diagram has a nonzero zeroth-order trace. The cancellation uses the sum of the four actual diagrams, not dummy-pair symmetrization or an SK time integral.

First-order matrices with both soft endpoints retained. For compact notation only, set

$$\mathcal{K}_j^{12} = \mathcal{K}_{j;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2), \qquad \mathcal{K}_j^{21} = \mathcal{K}_{j;\mathsf{a}_2\mathsf{a}_1}(k_L;\tau_2,\tau_1), \qquad E_v = E_{\mathsf{a}_v}(k_v,\tau_v). \tag{53}$$

The full first-order left integrand of 1234 is

$$\mathcal{I}_{12}^{(1)} = \mathsf{a}_1\mathsf{a}_2(-\tau_1)^{\mathrm{i}\tilde{\nu}-1}(-\tau_2)^{\mathrm{i}\tilde{\nu}-1}E_1E_2\Bigg\{ \underbrace{-\frac{1}{k_L}\boldsymbol{\sigma}\cdot\Big[\mathcal{K}_0^{12}\boldsymbol{\ell} + (p\partial_p - 1)\mathcal{K}_{0;\mathsf{a}_1\mathsf{a}_2}(p;\tau_1,\tau_2)\big|_{p=k_L}\,\mathbf{n}_L(\mathbf{n}_L\cdot\boldsymbol{\ell})\Big]}_{\text{hard-line correction}} + \underbrace{\frac{\mathrm{i}}{1+2\mathrm{i}\tilde{\nu}}[\tau_1\mathcal{K}_1^{12}\boldsymbol{\sigma}\cdot\mathbf{q} + \tau_2\mathcal{K}_1^{21}\boldsymbol{\sigma}\cdot\mathbf{K}]}_{\text{two separate soft-endpoint corrections}}\Bigg\}. \tag{54}$$

The Higgs factors in the first line are exact multiplicative factors. There is no external-mode derivative term.

To take each trace directly, write

$$V_{12}^{(1)} = -\frac{\boldsymbol{\sigma}\cdot\mathbf{w}_L^{12}}{k_L}, \quad V_{21}^{(1)} = -\frac{\boldsymbol{\sigma}\cdot\mathbf{w}_L^{21}}{k_L}, \quad V_{34}^{(1)} = -\frac{\boldsymbol{\sigma}\cdot\mathbf{w}_R^{34}}{k_R}, \quad V_{43}^{(1)} = -\frac{\boldsymbol{\sigma}\cdot\mathbf{w}_R^{43}}{k_R}. \tag{55}$$

The two left vectors are explicitly

$$\begin{aligned} \mathbf{w}_L^{12} &= \mathcal{K}_0^{12}\boldsymbol{\ell} + (p\partial_p - 1)\mathcal{K}_{0;\mathsf{a}_1\mathsf{a}_2}(p;\tau_1,\tau_2)\big|_{p=k_L}\,\mathbf{n}_L(\mathbf{n}_L\cdot\boldsymbol{\ell}) - \frac{\mathrm{i}k_L}{1+2\mathrm{i}\tilde{\nu}}[\tau_1\mathcal{K}_1^{12}\mathbf{q} + \tau_2\mathcal{K}_1^{21}\mathbf{K}], \\ \mathbf{w}_L^{21} &= \mathcal{K}_0^{12}\boldsymbol{\ell} + (p\partial_p - 1)\mathcal{K}_{0;\mathsf{a}_1\mathsf{a}_2}(p;\tau_1,\tau_2)\big|_{p=k_L}\,\mathbf{n}_L(\mathbf{n}_L\cdot\boldsymbol{\ell}) - \frac{\mathrm{i}k_L}{1+2\mathrm{i}\tilde{\nu}}[\tau_2\mathcal{K}_1^{21}\mathbf{q} + \tau_1\mathcal{K}_1^{12}\mathbf{K}]. \end{aligned} \tag{56}$$

The hard corrections agree by Eq. (50), while the coefficients of $\mathbf{q}$ and $\mathbf{K}$ are interchanged because those soft lines attach to different vertices in the second diagram. The right vectors are

$$(\mathbf{w}_R^{34}, \mathbf{w}_R^{43}) = (\mathbf{w}_L^{12}, \mathbf{w}_L^{21})\Big|_{(L,1,2,\mathbf{q},\mathbf{K})\to(R,3,4,\mathbf{K},\mathbf{q})}. \tag{57}$$

No relation $\mathcal{K}_1^{12} = \mathcal{K}_1^{21}$ is assumed.

Direct trace, followed by the diagram sum. For each of the four choices $A = 12, 21$ and $B = 34, 43$, the Pauli identity gives

$$\mathrm{tr}_2[V_A^{(1)}\sigma_K V_B^{(1)}\sigma_q] = \frac{2}{k_L k_R q K}\Big[2(\mathbf{w}_L^A\cdot\boldsymbol{\ell})(\mathbf{w}_R^B\cdot\boldsymbol{\ell}) - \tfrac{1}{2}(\mathbf{w}_L^A\cdot\mathbf{k}_s)(\mathbf{w}_R^B\cdot\mathbf{k}_s) - (\mathbf{w}_L^A\cdot\mathbf{w}_R^B)(\ell^2 - k_s^2/4)\Big]. \tag{58}$$

This evaluates the spin trace without combining the two soft endpoints of any single diagram. The scalar expression is bilinear in its two vectors, so the four results can now be added using their vector sums. In particular, the endpoint terms add as

$$[\tau_1\mathcal{K}_1^{12}\mathbf{q} + \tau_2\mathcal{K}_1^{21}\mathbf{K}] + [\tau_2\mathcal{K}_1^{21}\mathbf{q} + \tau_1\mathcal{K}_1^{12}\mathbf{K}] = 2(\tau_1\mathcal{K}_1^{12} + \tau_2\mathcal{K}_1^{21})\boldsymbol{\ell}. \tag{59}$$

This is the addition of two diagrams, with no factor 1/2 and no average over time or SK labels. Collect the scalar coefficients of the resulting vector sum:

$$\begin{aligned}
&\Pi^L_{1;\mathsf{a}_1\mathsf{a}_2} = \mathcal{K}_0^{12} - \frac{\mathrm{i}k_L}{1+2\mathrm{i}\tilde{\nu}}[\tau_1\mathcal{K}_1^{12} + \tau_2\mathcal{K}_1^{21}],\\
&\Pi^L_{2;\mathsf{a}_1\mathsf{a}_2} = (p\partial_p - 1)\mathcal{K}_{0;\mathsf{a}_1\mathsf{a}_2}(p;\tau_1,\tau_2)\big|_{p=k_L}, \qquad \Pi^R_{i;\mathsf{a}_3\mathsf{a}_4} = \Pi^L_{i;\mathsf{a}_1\mathsf{a}_2}\Big|_{(L,1,2)\to(R,3,4)}.
\end{aligned} \tag{60}$$

Both are still functions of their two times. Their associated vectors satisfy

$$\begin{aligned}
&\mathbf{v}_L = \Pi_1^L\boldsymbol{\ell} + \Pi_2^L\mathbf{n}_L(\mathbf{n}_L\cdot\boldsymbol{\ell}), && \mathbf{w}_L^{12} + \mathbf{w}_L^{21} = 2\mathbf{v}_L,\\
&\mathbf{v}_R = \Pi_1^R\boldsymbol{\ell} + \Pi_2^R\mathbf{n}_R(\mathbf{n}_R\cdot\boldsymbol{\ell}), && \mathbf{w}_R^{34} + \mathbf{w}_R^{43} = 2\mathbf{v}_R.
\end{aligned} \tag{61}$$

The sum of the four already evaluated traces is consequently

$$\sum_{\substack{A=12,21\\B=34,43}} \mathrm{tr}_2[V_A^{(1)}\sigma_K V_B^{(1)}\sigma_q] = \frac{8}{k_L k_R q K}\Big[2(\mathbf{v}_L\cdot\boldsymbol{\ell})(\mathbf{v}_R\cdot\boldsymbol{\ell}) - \tfrac{1}{2}(\mathbf{v}_L\cdot\mathbf{k}_s)(\mathbf{v}_R\cdot\mathbf{k}_s) - (\mathbf{v}_L\cdot\mathbf{v}_R)(\ell^2 - k_s^2/4)\Big]. \tag{62}$$

The factor 8 is 2 from the Pauli trace times 4 from the four diagrams. Restoring the common exact external weight, the complete leading same-branch integrand, including both oscillatory branches, is

$$\begin{aligned}
\mathcal{I} = \frac{8c^2}{k_L k_R}&\mathcal{E}_{\{\mathsf{a}\}}(\{\tau\};\{k\})(qK)^{2\mathrm{i}\tilde{\nu}-1}\\
&\times\Big[2(\mathbf{v}_L\cdot\boldsymbol{\ell})(\mathbf{v}_R\cdot\boldsymbol{\ell}) - \tfrac{1}{2}(\mathbf{v}_L\cdot\mathbf{k}_s)(\mathbf{v}_R\cdot\mathbf{k}_s) - (\mathbf{v}_L\cdot\mathbf{v}_R)(\ell^2 - k_s^2/4)\Big] + \text{c.c.}
\end{aligned} \tag{63}$$

The complex-conjugate term supplies the $-\mathrm{i}\tilde{\nu}$ branch in the complete SK sum, with all SK labels reversed under conjugation. Below we evaluate only the displayed $+\mathrm{i}\tilde{\nu}$ term and restore its conjugate once in the final result. All spinor indices have been contracted. The Higgs functions in $\mathcal{E}_{\{\mathsf{a}\}}$ still use the exact k_1, k_2, k_3, k_4 and take no part in the loop integral. The equal-energy limit needed for the leading time coefficient is taken only after that integral.

3.3 The scalar loop integral first

At fixed times and SK labels, the four $\Pi_i^{L,R}$ are independent of $\mathbf{q}$. The loop integral to be evaluated is therefore simply

$$\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}(qK)^{2\mathrm{i}\tilde{\nu}-1}\left[2(\mathbf{v}_L\cdot\boldsymbol{\ell})(\mathbf{v}_R\cdot\boldsymbol{\ell})-\tfrac{1}{2}(\mathbf{v}_L\cdot\mathbf{k}_s)(\mathbf{v}_R\cdot\mathbf{k}_s)-(\mathbf{v}_L\cdot\mathbf{v}_R)(\ell^2-k_s^2/4)\right]$$
$$=k_s^{5+4\mathrm{i}\tilde{\nu}}\sum_{i,j=1}^{2}\mathscr{L}_{ij}(\tilde{\nu};\theta_1,\theta_3,\phi_3)\Pi_i^L\Pi_j^R. \tag{64}$$

The factor $8/(k_Lk_R)$ from the trace is outside this definition. Choose

$$\mathbf{n}_L=(\sin\theta_1,0,\cos\theta_1),\quad \mathbf{n}_R=(\sin\theta_3\cos\phi_3,\sin\theta_3\sin\phi_3,\cos\theta_3),\quad \mathbf{n}_s=(0,0,1).$$

The loop integral has mass dimension $5+4\mathrm{i}\tilde{\nu}$. We evaluate it using the scalar convolution

$$f_0(A,B)=\frac{\Gamma(A+B-\frac{3}{2})\Gamma(\frac{3}{2}-A)\Gamma(\frac{3}{2}-B)}{\Gamma(A)\Gamma(B)\Gamma(3-A-B)}, \tag{65}$$
$$\int\frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}(q^2)^{-A}(K^2)^{-B}=\frac{k_s^{3-2A-2B}}{(4\pi)^{3/2}}f_0(A,B). \tag{66}$$

For the scalar numerator this identity follows by Schwinger parameters and a Gaussian momentum integral in the convergence domain $\Re A,\Re B<3/2$, $\Re(A+B)>3/2$. The shifted polynomial moments are defined by the same meromorphic continuation, with common regulator parameters before recombination. The physical soft-region integrals have UV power divergences when extended beyond $q\sim k_s$; this continuation selects their nonanalytic coefficient. UV-dependent polynomial contact terms belong to matching with the hard region and are excluded. No physical UV cutoff is sent to infinity term by term. The following reduction expresses the polynomial numerator as a finite sum of these scalar convolutions. The result depends on the two hard directions relative to $\mathbf{k}_s$ as well as on their mutual angle.

Reduction of the polynomial numerator. Write the numerator as $N=\sum_{i,j}\Pi_i^L\Pi_j^RN_{ij}$, and set $x=-\mathrm{i}\tilde{\nu}$ and $z=\widehat{\mathbf{q}}\cdot\widehat{\mathbf{k}}_s$. The angular moments are defined through the scalar convolution by

$$\mathcal{J}_r(m,n;x)=\frac{1}{2^r}\sum_{u+v+w=r}\frac{r!}{u!v!w!}(-1)^{v+w}f_0\left(x-\frac{m}{2}+\frac{r}{2}-v,x+\frac{n}{2}-u\right),\qquad r=0,1,2,3,4, \tag{67}$$

where the sum runs over nonnegative integers. This follows from $z=(K^2-q^2-k_s^2)/(2qk_s)$. Thus $K^2=q^2+k_s^2+2qk_sz$. Let an overline denote the average over the azimuth of $\mathbf{q}$ about $\mathbf{k}_s$, holding the external directions $\mathbf{n}_L$ and $\mathbf{n}_R$ fixed. Writing $c_1=\cos\theta_1$ and $c_3=\cos\theta_3$, the numerator is

$$\begin{aligned}\overline{N}_{11}&=M,\qquad \overline{N}_{12}=P+c_3^2Q,\qquad \overline{N}_{21}=P+c_1^2Q,\\ \overline{N}_{22}&=H_0+H_2(c_1^2+c_3^2)+H_3c_1^2c_3^2+H_B\sin(2\theta_1)\sin(2\theta_3)\cos\phi_3\\ &\quad+H_D\sin^2\theta_1\sin^2\theta_3\cos(2\phi_3).\end{aligned} \tag{68}$$

Using $\boldsymbol{\ell} = \mathbf{q} + \mathbf{k}_s/2$ and performing this azimuthal average gives the following polynomials:

$$
\begin{aligned}
M &= q^4 + 2k_s q^3 z + \frac{k_s^2 q^2}{4}(3 + 2z^2) + \frac{k_s^3 q z}{4}, \\
P &= \frac{q^4}{2}(1 - z^2) + \frac{k_s q^3}{2}(z - z^3) + \frac{k_s^2 q^2}{4}(1 - z^2), \\
Q &= \frac{q^4}{2}(3z^2 - 1) + \frac{k_s q^3}{2}(3z^3 + z) + \frac{5}{4}k_s^2 q^2 z^2 + \frac{k_s^3 q z}{4}, \\
H_0 &= \frac{q^4}{4}(2z^4 - 3z^2 + 1) + \frac{k_s q^3}{4}(z^3 - z), \\
H_2 &= \frac{q^4}{4}(-6z^4 + 7z^2 - 1) + \frac{5}{4}k_s q^3(z - z^3) + \frac{k_s^2 q^2}{4}(1 - z^2), \\
H_3 &= \frac{q^4}{4}(18z^4 - 15z^2 + 1) + \frac{k_s q^3}{4}(21z^3 - 13z) + \frac{k_s^2 q^2}{4}(8z^2 - 3) + \frac{k_s^3 q z}{4}, \\
H_B &= \frac{q^4}{8}(-8z^4 + 7z^2 - 1) + \frac{k_s q^3}{8}(5z - 9z^3) + \frac{k_s^2 q^2}{16}(2 - 7z^2) - \frac{k_s^3 q z}{16}, \\
H_D &= \frac{q^4}{4}(z^4 - z^2) + \frac{k_s q^3}{4}(z^3 - z).
\end{aligned}
\tag{69}
$$

The common weight is $(qK)^{-2x-1}$. Each monomial therefore reduces directly to the functions defined in (67):

$$
\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}(qK)^{-2x-1}k_s^{4-a}q^a z^r = \frac{k_s^{5-4x}}{(4\pi)^{3/2}}\,\mathcal{J}_r(a-1,1;x). \tag{70}
$$

There is no factor 4^{2x} here, since the soft weight contains qK rather than $(q/2)(K/2)$. For brevity, the common argument x of $\mathcal{J}_r(m,n;x)$ is suppressed below. Define

$$
\begin{aligned}
\mathcal{U}_M &= \mathcal{J}_0(3,1) + 2\mathcal{J}_1(2,1) + \frac{1}{2}\mathcal{J}_2(1,1) + \frac{3}{4}\mathcal{J}_0(1,1) + \frac{1}{4}\mathcal{J}_1(0,1), \\
\mathcal{U}_P &= \frac{1}{2}[\mathcal{J}_0(3,1) - \mathcal{J}_2(3,1)] + \frac{1}{2}[\mathcal{J}_1(2,1) - \mathcal{J}_3(2,1)] + \frac{1}{4}[\mathcal{J}_0(1,1) - \mathcal{J}_2(1,1)], \\
\mathcal{U}_Q &= \frac{1}{2}[3\mathcal{J}_2(3,1) - \mathcal{J}_0(3,1)] + \frac{1}{2}[3\mathcal{J}_3(2,1) + \mathcal{J}_1(2,1)] + \frac{5}{4}\mathcal{J}_2(1,1) + \frac{1}{4}\mathcal{J}_1(0,1).
\end{aligned}
\tag{71}
$$

The five angular coefficients are

$$
\begin{aligned}
\mathcal{U}_0 &= \frac{1}{4}[2\mathcal{J}_4(3,1) - 3\mathcal{J}_2(3,1) + \mathcal{J}_0(3,1) + \mathcal{J}_3(2,1) - \mathcal{J}_1(2,1)], \\
\mathcal{U}_2 &= \frac{1}{4}[-6\mathcal{J}_4(3,1) + 7\mathcal{J}_2(3,1) - \mathcal{J}_0(3,1)] + \frac{5}{4}[\mathcal{J}_1(2,1) - \mathcal{J}_3(2,1)] \\
&\quad + \frac{1}{4}[\mathcal{J}_0(1,1) - \mathcal{J}_2(1,1)], \\
\mathcal{U}_3 &= \frac{1}{4}[18\mathcal{J}_4(3,1) - 15\mathcal{J}_2(3,1) + \mathcal{J}_0(3,1)] + \frac{1}{4}[21\mathcal{J}_3(2,1) - 13\mathcal{J}_1(2,1)] \\
&\quad + 2\mathcal{J}_2(1,1) - \frac{3}{4}\mathcal{J}_0(1,1) + \frac{1}{4}\mathcal{J}_1(0,1), \\
\mathcal{U}_B &= \frac{1}{8}[-8\mathcal{J}_4(3,1) + 7\mathcal{J}_2(3,1) - \mathcal{J}_0(3,1)] + \frac{1}{8}[5\mathcal{J}_1(2,1) - 9\mathcal{J}_3(2,1)] \\
&\quad + \frac{1}{16}[2\mathcal{J}_0(1,1) - 7\mathcal{J}_2(1,1) - \mathcal{J}_1(0,1)], \\
\mathcal{U}_D &= \frac{1}{4}[\mathcal{J}_4(3,1) - \mathcal{J}_2(3,1) + \mathcal{J}_3(2,1) - \mathcal{J}_1(2,1)].
\end{aligned}
\tag{72}
$$

The unreduced loop coefficients are consequently

$$
\begin{aligned}
(4\pi)^{3/2}\mathscr{L}_{11} &= \mathcal{U}_M, \qquad (4\pi)^{3/2}\mathscr{L}_{12} = \mathcal{U}_P + c_3^2\mathcal{U}_Q, \qquad (4\pi)^{3/2}\mathscr{L}_{21} = \mathcal{U}_P + c_1^2\mathcal{U}_Q,\\
(4\pi)^{3/2}\mathscr{L}_{22} &= \mathcal{U}_0 + \mathcal{U}_2(c_1^2+c_3^2) + \mathcal{U}_3 c_1^2 c_3^2\\
&\quad + \mathcal{U}_B \sin(2\theta_1)\sin(2\theta_3)\cos\phi_3\\
&\quad + \mathcal{U}_D \sin^2\theta_1 \sin^2\theta_3 \cos(2\phi_3).
\end{aligned}
\tag{73}
$$

Applying the finite sum (67) and the Gamma recurrence relation gives $\mathcal{U}_Q = 0$, $\mathcal{U}_M = 3\mathcal{U}_P$, and $\mathcal{U}_3 = -3\mathcal{U}_2$. In particular, the angular dependence of $\mathscr{L}_{12}$ and $\mathscr{L}_{21}$ cancels only after the loop integration. The common Gamma factor is

$$
\mathcal{G}_{\rm top}(\tilde\nu) = \frac{\Gamma(-\frac{1}{2}-2\mathrm{i}\tilde\nu)\Gamma^2(1+\mathrm{i}\tilde\nu)}{(4\pi)^{3/2}\Gamma(2+2\mathrm{i}\tilde\nu)\Gamma^2(\frac{1}{2}-\mathrm{i}\tilde\nu)}.
\tag{74}
$$

The fully reduced coefficients are

$$
\begin{aligned}
\mathscr{L}_{11} &= 3\mathscr{L}_{12} = 3\mathscr{L}_{21},\\
\mathscr{L}_{12} &= -\mathcal{G}_{\rm top}\,\frac{(1+\mathrm{i}\tilde\nu)(1+2\mathrm{i}\tilde\nu)}{4(3+2\mathrm{i}\tilde\nu)(5+4\mathrm{i}\tilde\nu)(3+4\mathrm{i}\tilde\nu)},\\
\mathscr{L}_{22} &= -\frac{\mathcal{G}_{\rm top}(1+\mathrm{i}\tilde\nu)}{4(5+2\mathrm{i}\tilde\nu)(3+2\mathrm{i}\tilde\nu)(5+4\mathrm{i}\tilde\nu)(3+4\mathrm{i}\tilde\nu)}\\
&\quad\times\Big\{(2+\mathrm{i}\tilde\nu)(1+2\mathrm{i}\tilde\nu) - (1+\mathrm{i}\tilde\nu)(1+2\mathrm{i}\tilde\nu)[c_1^2+c_3^2-3c_1^2c_3^2]\\
&\quad - \frac{1}{2}(1+2\mathrm{i}\tilde\nu)(3+2\mathrm{i}\tilde\nu)\sin(2\theta_1)\sin(2\theta_3)\cos\phi_3\\
&\quad + (2+\mathrm{i}\tilde\nu)(3+2\mathrm{i}\tilde\nu)\sin^2\theta_1\sin^2\theta_3\cos(2\phi_3)\Big\}.
\end{aligned}
\tag{75}
$$

The $\eta = -1$ branch follows by complex conjugation. The $\mathcal{J}_r$ are evaluated by analytic continuation. This extracts the nonanalytic soft coefficient; polynomial contact terms from matching to the hard region are not included. No time integral was used to obtain these coefficients. The left and right time kernels remain independent throughout the loop reduction.

3.4 Left and right time integrals from the Whittaker seed

After the loop integral, the exact external factors enter only the two scalar time integrals. Before taking the collapsed limit these are

$$
\begin{aligned}
\mathcal{T}_i^L(k_1,k_2;k_L) &= \sum_{\mathsf{a}_1,\mathsf{a}_2=\pm} \mathsf{a}_1\mathsf{a}_2 \int_{-\infty}^0 \mathrm{d}\tau_1\mathrm{d}\tau_2\,(-\tau_1)^{\mathrm{i}\tilde\nu-1}(-\tau_2)^{\mathrm{i}\tilde\nu-1} E_{\mathsf{a}_1}(k_1,\tau_1)E_{\mathsf{a}_2}(k_2,\tau_2)\Pi^L_{i;\mathsf{a}_1\mathsf{a}_2},\\
\mathcal{T}_j^R(k_3,k_4;k_R) &= \sum_{\mathsf{a}_3,\mathsf{a}_4=\pm} \mathsf{a}_3\mathsf{a}_4 \int_{-\infty}^0 \mathrm{d}\tau_3\mathrm{d}\tau_4\,(-\tau_3)^{\mathrm{i}\tilde\nu-1}(-\tau_4)^{\mathrm{i}\tilde\nu-1} E_{\mathsf{a}_3}(k_3,\tau_3)E_{\mathsf{a}_4}(k_4,\tau_4)\Pi^R_{j;\mathsf{a}_3\mathsf{a}_4}.
\end{aligned}
\tag{76}
$$

Here $i,j = 1,2$. Their product multiplies $k_s^{5+4\mathrm{i}\tilde\nu}\mathscr{L}_{ij}$ from the completed loop integral. For the leading collapsed signal, take the finite equal-energy limit of these time coefficients, after combining the SK branches:

$$
\mathcal{T}_i^L(k_L,k_L;k_L) = k_L^{-2\mathrm{i}\tilde\nu}\mathcal{P}_i(\tilde\nu), \qquad \mathcal{T}_j^R(k_R,k_R;k_R) = k_R^{-2\mathrm{i}\tilde\nu}\mathcal{P}_j(\tilde\nu).
\tag{77}
$$

Corrections to this limit affect higher soft orders. No Taylor expansion of a Higgs mode is needed in the loop calculation. A common endpoint regulator is retained until the complete SK sum has been combined. With $x = -k_L\tau_1$, $y = -k_L\tau_2$ and $r = p/k_L$, introduce the same two scalar moments used for the mixed branch:

$$\mathcal{H}_j(\alpha,\beta;r) = \sum_{\mathsf{a},\mathsf{b}=\pm} \mathsf{ab} \int_0^\infty \mathrm{d}x\mathrm{d}y\, x^{\alpha-1}y^{\beta-1}(1+\mathrm{i}\mathsf{a}x)(1+\mathrm{i}\mathsf{b}y)e^{-\mathrm{i}\mathsf{a}x-\mathrm{i}\mathsf{b}y}\mathcal{K}_{j;\mathsf{ab}}(r;x,y), \qquad j=0,1. \tag{78}$$

Here $\mathcal{K}_j(r;x,y)$ means Eq. (41) at $X = rx$, $Y = ry$. The two endpoint terms are kept separately up to this stage. Their Whittaker seed evaluations give

$$\begin{aligned}
&\sum_{\mathsf{a},\mathsf{b}} \mathsf{ab} \int_0^\infty \mathrm{d}x\mathrm{d}y\, x^{\mathrm{i}\tilde\nu-1}y^{\mathrm{i}\tilde\nu-1}(1+\mathrm{i}\mathsf{a}x)(1+\mathrm{i}\mathsf{b}y)e^{-\mathrm{i}\mathsf{a}x-\mathrm{i}\mathsf{b}y}\, x\mathcal{K}_{1;\mathsf{ab}}(1;x,y) = \mathcal{H}_1(1+\mathrm{i}\tilde\nu,\mathrm{i}\tilde\nu;1),\\
&\sum_{\mathsf{a},\mathsf{b}} \mathsf{ab} \int_0^\infty \mathrm{d}x\mathrm{d}y\, x^{\mathrm{i}\tilde\nu-1}y^{\mathrm{i}\tilde\nu-1}(1+\mathrm{i}\mathsf{a}x)(1+\mathrm{i}\mathsf{b}y)e^{-\mathrm{i}\mathsf{a}x-\mathrm{i}\mathsf{b}y}\, y\mathcal{K}_{1;\mathsf{ba}}(1;y,x) = \mathcal{H}_1(1+\mathrm{i}\tilde\nu,\mathrm{i}\tilde\nu;1).
\end{aligned} \tag{79}$$

The second uses the seed with its two arguments and SK labels in the reversed order. These are equal values of two completed time integrals, not a symmetrization of the unintegrated fermion factor. Adding them gives

$$\begin{aligned}
\mathcal{P}_1(\tilde\nu) &= \mathcal{H}_0(\mathrm{i}\tilde\nu,\mathrm{i}\tilde\nu;1) + \frac{2\mathrm{i}}{1+2\mathrm{i}\tilde\nu}\mathcal{H}_1(1+\mathrm{i}\tilde\nu,\mathrm{i}\tilde\nu;1),\\
\mathcal{P}_2(\tilde\nu) &= (\partial_r - 1)\mathcal{H}_0(\mathrm{i}\tilde\nu,\mathrm{i}\tilde\nu;r)\big|_{r=1}\,.
\end{aligned} \tag{80}$$

The first coefficient includes both the hard-direction expansion and the two soft endpoint descendants; the second combines the hard-magnitude derivative with the longitudinal direction correction. The same functions apply on the right. Combining the already evaluated loop with the two time integrals gives

$$\mathcal{G}^{\mathrm{top}}\Big|_{s,\mathrm{clock}} = -\frac{N_c g_t^4 H^4}{16\prod_i k_i^3}\left[\frac{8c^2k_s^{5+4\mathrm{i}\tilde\nu}}{(k_Lk_R)^{1+2\mathrm{i}\tilde\nu}}\sum_{i,j=1}^{2}\mathscr{L}_{ij}\mathcal{P}_i\mathcal{P}_j + \mathrm{c.c.}\right]. \tag{81}$$

It remains to express the scalar time moments through the existing Whittaker integral seed; no spin trace or momentum integral remains.

Seed normalization. We keep the scalar propagator normalization in Eq. (17). With $p = rk_L$, $x = -k_L\tau_1$ and $y = -k_L\tau_2$, write its dimensionless version as

$$D^{\mathrm{sc},\kappa}_{\mathsf{ab}}(r;x,y) \equiv k_L D^{\mathrm{sc},\kappa}_{\mathsf{ab}}(p;-x/k_L,-y/k_L).$$

The two hard coefficients in Eq. (41) then contain $A_x = x\partial_x - \mathrm{i}rx - 1/2$ and $R_x = -x\partial_x - \mathrm{i}rx + 1/2$, with common prefactor $\mathrm{i}/(2\sqrt{xy})$. Define their scalar time moments by

$$\mathcal{J}^{\kappa;P,Q}_{\mathsf{ab}}(r) = \int_0^\infty \mathrm{d}x\mathrm{d}y\, x^P y^Q e^{-\mathrm{i}\mathsf{a}x-\mathrm{i}\mathsf{b}y} D^{\mathrm{sc},\kappa}_{\mathsf{ab}}(r;x,y) = -\mathsf{ab}\, r^{-3-P-Q}\mathscr{J}^{\kappa;P,Q}_{\mathsf{ab}}(u,u), \quad u = \frac{2r}{1+r}. \tag{82}$$

Here $\mathscr{J}$ is precisely the Whittaker seed of Ref. [3], Eq. (11), with mass index $\tilde\nu$ and $\mathrm{i}h\tilde\mu = \kappa$. The phase $e^{\mathrm{i}\pi\kappa}$ is already in $D^{\mathrm{sc},\kappa}$; the seed definition includes $-\mathsf{ab}$. There is no Hankel mass-index shift.

Removing the time derivatives. Expand the external factors as

$$(1+\mathsf{i}\mathsf{a}x)(1+\mathsf{i}\mathsf{b}y)=\sum_{j,l=0}^{1}(\mathsf{i}\mathsf{a})^j(\mathsf{i}\mathsf{b})^l x^j y^l.$$

For each term put $P=\alpha-3/2+j$, $Q=\beta-3/2+l$. Integration by parts turns $x\partial_x$ into $-(P+1)+\mathsf{i}\mathsf{a}x$. Consequently

$$\mathcal{H}_0(\alpha,\beta;r)=\frac{\mathrm{i}}{2}\sum_{\mathsf{a},\mathsf{b}=\pm}\sum_{j,l=0}^{1}\mathsf{a}\mathsf{b}(\mathsf{i}\mathsf{a})^j(\mathsf{i}\mathsf{b})^l\Big[(P+\tfrac{3}{2}+\mathrm{i}\tilde{\nu})(\mathcal{J}^{-;P,Q}_{\mathsf{a}\mathsf{b}}-\mathcal{J}^{+;P,Q}_{\mathsf{a}\mathsf{b}}) + \mathrm{i}(\mathsf{a}-r)\mathcal{J}^{+;P+1,Q}_{\mathsf{a}\mathsf{b}}-\mathrm{i}(\mathsf{a}+r)\mathcal{J}^{-;P+1,Q}_{\mathsf{a}\mathsf{b}}\Big], \tag{83}$$

$$\mathcal{H}_1(\alpha,\beta;r)=\frac{\mathrm{i}}{2}\sum_{\mathsf{a},\mathsf{b}=\pm}\sum_{j,l=0}^{1}\mathsf{a}\mathsf{b}(\mathsf{i}\mathsf{a})^j(\mathsf{i}\mathsf{b})^l\Big[(P+\tfrac{3}{2}-\mathrm{i}\tilde{\nu})(\mathcal{J}^{-;P,Q}_{\mathsf{a}\mathsf{b}}+\mathcal{J}^{+;P,Q}_{\mathsf{a}\mathsf{b}}) - \mathrm{i}(\mathsf{a}-r)\mathcal{J}^{+;P+1,Q}_{\mathsf{a}\mathsf{b}}-\mathrm{i}(\mathsf{a}+r)\mathcal{J}^{-;P+1,Q}_{\mathsf{a}\mathsf{b}}\Big]. \tag{84}$$

All moments have argument r, and their superscripts $+$ and $-$ mean $\kappa=\pm1/2$. These two formulas apply both to equal and opposite soft branches. They use only the existing scalar Whittaker seed and integer shifts of its time powers.

Integration by parts is first performed with a common endpoint regulator in a domain where its boundary terms vanish, and with the BD early-time prescription. The complete seed combination is then continued to $\delta\to0$. Boundary terms or poles of separate SK terms must not be dropped before this combination.

Closed-form seed values. At $r=1$ the needed moments use $\mathscr{S}^{\kappa;P,Q}_{\mathsf{a}\mathsf{b}}=\mathscr{I}^{\kappa;P,Q}_{\mathsf{a}\mathsf{b}}(1,1)$. For compactness $\Gamma[z_1,\dots,z_n]$ denotes the product of Gamma functions. The results of Ref. [3], Eqs. (130)–(131), become

$$\mathscr{S}^{\kappa;P,Q}_{\mathsf{a},-\mathsf{a}}=\frac{e^{-\mathsf{i}\mathsf{a}\pi(P-Q)/2+\mathrm{i}\pi\kappa}}{2^{3+P+Q}}\frac{\Gamma[\frac{3}{2}+P-\mathrm{i}\tilde{\nu},\frac{3}{2}+P+\mathrm{i}\tilde{\nu},\frac{3}{2}+Q-\mathrm{i}\tilde{\nu},\frac{3}{2}+Q+\mathrm{i}\tilde{\nu}]}{\Gamma[2+P+\mathsf{a}\kappa,2+Q-\mathsf{a}\kappa]}, \tag{85}$$

$$\begin{aligned}\mathscr{S}^{\kappa;P,Q}_{\mathsf{a}\mathsf{a}}=&\frac{-\mathsf{a}\mathrm{i}\,e^{-\mathsf{i}\mathsf{a}\pi(P+Q)/2}}{2^{3+P+Q}}\Gamma[\tfrac{3}{2}+P+\mathsf{a}\mathrm{i}\tilde{\nu},\tfrac{3}{2}+Q+\mathsf{a}\mathrm{i}\tilde{\nu}]\\&\times\Bigg\{\frac{e^{2\mathrm{i}\pi\kappa}+e^{-2\pi\tilde{\nu}}}{2\pi}\frac{\Gamma[\frac{3}{2}+P-\mathsf{a}\mathrm{i}\tilde{\nu},\frac{3}{2}+Q-\mathsf{a}\mathrm{i}\tilde{\nu},\frac{1}{2}+\mathsf{a}\kappa-\mathrm{i}\tilde{\nu},\frac{1}{2}+\mathsf{a}\kappa+\mathrm{i}\tilde{\nu}]}{\Gamma[2+P+\mathsf{a}\kappa,2+Q+\mathsf{a}\kappa]}\\&+\mathsf{a}\mathrm{i}\Gamma(3+P+Q)\,{}_3\widetilde{F}_2\left(\begin{matrix}3+P+Q,\ \frac{1}{2}-\mathsf{a}\kappa+\mathsf{a}\mathrm{i}\tilde{\nu},\ 1\\ \frac{5}{2}+P+\mathsf{a}\mathrm{i}\tilde{\nu},\ \frac{5}{2}+Q+\mathsf{a}\mathrm{i}\tilde{\nu}\end{matrix};1\right)\Bigg\}.\end{aligned} \tag{86}$$

The regularized function is

$${}_3\widetilde{F}_2(A,B,C;D,E;1)=\frac{{}_3F_2(A,B,C;D,E;1)}{\Gamma(D)\Gamma(E)}. \tag{87}$$

At $\kappa=\pm1/2$ some displayed series have zero real convergence balance. The expression denotes analytic continuation, not direct summation of a divergent series. A useful convergent representation is

$${}_3\widetilde{F}_2(A,B,C;D,E;1)=\frac{\Gamma(s)}{\Gamma(E-C)\Gamma(D)\Gamma(s+C)}{}_3F_2(D-A,D-B,C;D,s+C;1),. \tag{88}$$

Here $s = D+E-A-B-C$. Permuting upper and lower parameters allows a representation with $\mathrm{Re}(E-C) > 0$. This identity follows by applying the Euler transformation inside the beta-integral representation and continuing analytically.

Hard-momentum derivative. The Whittaker bootstrap equation also determines the derivative needed for $\mathcal{P}_2$. Define

$$T^{P,Q}_{\mathsf{ab}} = \delta_{\mathsf{ab}}\, e^{-\mathrm{i}\mathsf{a}\pi(P+Q)/2} 2^{-3-P-Q}\Gamma(3+P+Q).$$

At $u_1 = u_2 = 1$ its second-derivative coefficient vanishes, giving

$$\begin{aligned}\mathscr{R}^{\kappa;P,Q}_{\mathsf{ab}} &\equiv \frac{\mathrm{d}}{\mathrm{d}r}\mathscr{I}^{\kappa;P,Q}_{\mathsf{ab}}(u(r),u(r))\Big|_{r=1} \\ &= \frac{1}{2}\left[\frac{\tilde\nu^2+(P+\frac{3}{2})^2}{2+P+\mathsf{a}\kappa}+\frac{\tilde\nu^2+(Q+\frac{3}{2})^2}{2+Q+\mathsf{b}\kappa}\right]\mathscr{S}^{\kappa;P,Q}_{\mathsf{ab}}+\frac{1}{2}\left[\frac{1}{2+P+\mathsf{a}\kappa}+\frac{1}{2+Q+\mathsf{b}\kappa}\right]T^{P,Q}_{\mathsf{ab}},\end{aligned} \tag{89}$$

$$\partial_r \mathcal{J}^{\kappa;P,Q}_{\mathsf{ab}}\Big|_{r=1} = -\mathsf{ab}\left[\mathscr{R}^{\kappa;P,Q}_{\mathsf{ab}} - (3+P+Q)\mathscr{S}^{\kappa;P,Q}_{\mathsf{ab}}\right]. \tag{90}$$

Differentiate Eq. (83), including its explicit $\mathsf{a}\pm r$ factors, and use these moment derivatives. The required coefficients are therefore

$$\begin{aligned}\mathcal{P}_1 &= \lim_{\delta\to 0}\left[\mathcal{H}_0(\mathrm{i}\tilde\nu+\delta,\mathrm{i}\tilde\nu+\delta;1)+\frac{2\mathrm{i}}{1+2\mathrm{i}\tilde\nu}\mathcal{H}_1(1+\mathrm{i}\tilde\nu+\delta,\mathrm{i}\tilde\nu+\delta;1)\right], \\ \mathcal{P}_2 &= \lim_{\delta\to 0}\,(\partial_r-1)\mathcal{H}_0(\mathrm{i}\tilde\nu+\delta,\mathrm{i}\tilde\nu+\delta;r)|_{r=1}\,.\end{aligned} \tag{91}$$

Thus both hard subgraphs are calculated entirely from Whittaker scalar seed values; no new time integral is left.

Finite evaluation. Combining the complete SK sums and simplifying the hypergeometric functions gives the following finite representation. For this subsection only set $z = \mathrm{i}\tilde\nu$, and define the single symmetric family

$$\mathcal{Q}_{jl} = \frac{2^{-1-2z-j-l}\Gamma(1+2z+j+l)}{(1+2z+j)(1+2z+l)}\,{}_3F_2\left(\begin{matrix}1+2z+j+l,\ 1+z,\ 1\\ 2+2z+j,\ 2+2z+l\end{matrix};1\right). \tag{92}$$

Only $\mathcal{Q}_{00},\mathcal{Q}_{01},\mathcal{Q}_{11},\mathcal{Q}_{02},\mathcal{Q}_{12}$ are needed. Also abbreviate the finite endpoint contribution by

$$\mathcal{H}_{\mathrm{end}} = \pi^2 2^{-1-2z}\frac{\Gamma(1+2z)^2}{\Gamma(1+z)^2}. \tag{93}$$

The fully summed hard seeds and momentum derivative are

$$\begin{aligned}\mathcal{H}_0(z,z;1) &= 2\cosh(\pi\tilde\nu)(\mathcal{Q}_{00}+2\mathcal{Q}_{01}+\mathcal{Q}_{11})-\mathcal{H}_{\mathrm{end}}, \\ \mathcal{H}_1(1+z,z;1) &= -2\mathrm{i}\cosh(\pi\tilde\nu)(\mathcal{Q}_{02}+\mathcal{Q}_{12}),\end{aligned} \tag{94}$$

$$\begin{aligned}\mathcal{P}_2 = 2\cosh(\pi\tilde\nu)\sum_{j,l=0}^{1}&\left[\frac{j(1+2z+j)}{2(1+z+j)}+\frac{l(1+2z+l)}{2(1+z+l)}-(1+2z+j+l)\right]\mathcal{Q}_{jl} \\ &+(1+2z)\mathcal{H}_{\mathrm{end}}+2^{-1-2z}\cosh(\pi\tilde\nu)\Gamma(1+2z)\frac{4+5z+2z^2}{1+z}.\end{aligned} \tag{95}$$

Together with $\mathcal{P}_1$ in Eq. (80), these are finite, convergent special-function expressions for the complete hard coefficients. No raw mode-power-series truncation or numerical regulator extrapolation is required for the oscillatory signal.

The late-time cancellation can also be seen before integration. For either hard coefficient $\mathcal{K}_j$, with $E_{\mathsf{a}}(x) = (1 + \mathsf{i}\mathsf{a}x)e^{-\mathsf{i}\mathsf{a}x}$, the SK sum for $x < y$ is

$$\sum_{\mathsf{a},\mathsf{b}} \mathsf{a}\mathsf{b} E_{\mathsf{a}}(x) E_{\mathsf{b}}(y) \mathcal{K}_{j;\mathsf{a}\mathsf{b}}(x,y) = [E_+(x) - E_-(x)][E_+(y)\mathcal{K}_{j;-+}(x,y) - E_-(y)\mathcal{K}_{j;+-}(x,y)]. \tag{96}$$

Since $E_+(x) - E_-(x) = -2\mathrm{i}x^3/3 + O(x^5)$, the common late-time endpoint is integrable in the hard blocks used here. The early-time BD prescription is still necessary; this identity does not justify integrating undamped oscillations numerically on a truncated interval.

For reference, direct evaluation of Eqs. (94) and (95) gives

$\tilde{\nu}$	$\mathcal{P}_1$	$\mathcal{P}_2$
0.20	$-1.030411081 - 1.109020094\mathrm{i}$	$3.163771501 + 0.704629904\mathrm{i}$
0.73	$-1.598130708 - 2.346219694\mathrm{i}$	$1.923885009 + 1.958596831\mathrm{i}$
1.00	$-1.325125325 - 2.665004109\mathrm{i}$	$1.137292037 + 2.131408832\mathrm{i}$
2.00	$1.901463897 - 2.535666708\mathrm{i}$	$-1.119241948 + 1.383844867\mathrm{i}$
5.00	$3.921820606 + 1.458261456\mathrm{i}$	$-2.047325876 - 0.690483918\mathrm{i}$

The branch-resolved common-regulator limit is checked at $\tilde{\nu} = 0.73$, and the finite coefficients are compared at all five masses. The implementation and reproducible checks are listed in Section 5.

3.5 Total oscillatory trispectrum

Rewriting Eq. (81) in the usual clock ratio gives the leading nonzero oscillatory term, summed over the four s-cut boxes:

$$\begin{aligned}\langle \delta h_{\mathbf{k}_1} \delta h_{\mathbf{k}_2} \delta h_{\mathbf{k}_3} \delta h_{\mathbf{k}_4} \rangle'|_{s,\mathrm{clock}} = &-\frac{N_c g_t^4 H^4 k_s^5}{16\, k_1^3 k_2^3 k_3^3 k_4^3 k_L k_R} \\ &\times \left[\left(\frac{k_s^2}{4 k_{12} k_{34}} \right)^{2\mathrm{i}\tilde{\nu}} C^{\mathrm{Box}}(\tilde{\nu}; \theta_1, \theta_3, \phi_3) + \mathrm{c.c.} \right].\end{aligned} \tag{97}$$

Here C^{Box} denotes the top-quark box coefficient. Using the same angular basis as the SMS manuscript, write

$$\begin{aligned}C^{\mathrm{Box}} = &\, A_1 + A_2(\cos^2\theta_1 + \cos^2\theta_3) + A_3 \cos^2\theta_1 \cos^2\theta_3 \\ &+ B \sin(2\theta_1)\sin(2\theta_3)\cos\phi_3 + D \sin^2\theta_1 \sin^2\theta_3 \cos(2\phi_3).\end{aligned} \tag{98}$$

The five functions A_1, A_2, A_3, B, D depend only on $\tilde{\nu}$; D here is an angular coefficient, not a propagator. All five coefficients share one soft-bubble normalization, including the two soft-mode coefficients c^2, the factor 4 from the two Bose-summed hard blocks, the factor 2 from the Pauli trace, and the phase from the chosen momentum ratio:

$$\mathcal{B}(\tilde{\nu}) = \frac{2^{1+4\mathrm{i}\tilde{\nu}} \Gamma(\frac{1}{2} - \mathrm{i}\tilde{\nu})^4 f_0(\rho,\rho)}{32\pi^2 (4\pi)^{3/2} (\rho-3)(\rho-2)(4\rho-7)(4\rho-5)}, \qquad \rho = \tfrac{1}{2} - \mathrm{i}\tilde{\nu}. \tag{99}$$

In particular, the natural hard scaling is $(k_s^2/(k_L k_R))^{2i\tilde{\nu}}$. Since $4k_{12}k_{34} = 16k_L k_R$ at this order, using the displayed ratio transfers $16^{2i\tilde{\nu}}$ into the coefficient. Thus $8c^2\,16^{2i\tilde{\nu}} = 2^{1+4i\tilde{\nu}}\Gamma(1/2-i\tilde{\nu})^4/\pi^2$, which fixes both the phase and the overall factor in Eq. (99). Combining the bubble with the two hard coefficients of Eq. (80) gives the final mass dependence:

$$A_1 = \mathcal{B}(\rho-1)(2\rho-3)\Big[4(\rho-3)(3\mathcal{P}_1^2+2\mathcal{P}_1\mathcal{P}_2)+(2\rho-5)\mathcal{P}_2^2\Big], \tag{100}$$

$$A_2 = -\mathcal{B}\mathcal{P}_2^2(\rho-1)(2\rho-3)^2, \qquad A_3 = -3A_2, \tag{101}$$

$$B = -2\mathcal{B}\mathcal{P}_2^2(\rho-1)(\rho-2)(2\rho-3), \tag{102}$$

$$D = \mathcal{B}\mathcal{P}_2^2(\rho-2)(2\rho-5)(2\rho-3). \tag{103}$$

Thus the answer requires only two hard functions $\mathcal{P}_1, \mathcal{P}_2$ and one explicit bubble factor $\mathcal{B}$. The coefficients are complex and include the clock phase. The full answer is even under either Higgs-pair exchange and under parity. Its engineering dimension is -5, as required for the primed four-point function. The powers of H follow from four rescaled Weyl Yukawa vertices and four external Higgs contractions; no extra factor of H or a second color/orientation multiplicity is implicit. For fixed $\tilde{\nu} > 0$ these are the leading double-soft nonanalytic terms; higher soft powers and analytic contact terms are not displayed. With the centered hard momenta the next clock terms begin at $k_s^{7\pm 4i\tilde{\nu}}$, and the next mixed terms at k_s^5. Thus the generic relative error within each branch is $O(k_s^2/k_L^2, k_s^2/k_R^2, k_s^2/(k_L k_R))$; it is not a relative-error bound at a zero of the displayed angular coefficient.

4 Nonoscillatory contribution

4.1 Left and right subgraphs

Taking opposite soft branches gives a nonoscillatory k_s^3 term. For $\eta = -1$, the leading terms in Eqs. (26)–(28) are represented by

$$L_{\mathrm{L}}^- = \mathbf{1}_2, \qquad L_{\overline{\mathrm{L}}}^- = -\boldsymbol{\epsilon}, \qquad U_{\mathrm{L}}^- = \boldsymbol{\epsilon}, \qquad U_{\overline{\mathrm{L}}}^- = -\mathbf{1}_2,$$

with scalar prefactor $+c^*/2$, compared with $-c/2$ for $\eta = +1$. Use U_r^- at one soft endpoint and L_s^+ at the other in Eq. (37), and interchange $+,-$ at the right end. The superscripts $-+$ and $+-$ below label the two nonlocal mass branches at the endpoints; the SK labels remain on the hard propagator. At zeroth order their explicit four-type sums give

$$V_L^{-+,(0)} = -\frac{1}{2}\sum_{r,s} U_r^- J_r D^{rs} J_s L_s^+ = \mathcal{K}_{1;\mathsf{a}_1\mathsf{a}_2}(\tau_1,\tau_2)\mathbf{1}_2,$$

$$V_R^{+-,(0)} = -\frac{1}{2}\sum_{r,s} U_r^+ J_r D^{rs} J_s L_s^- = \mathcal{K}_{1;\mathsf{a}_4\mathsf{a}_3}(\tau_4,\tau_3)\mathbf{1}_2. \tag{104}$$

The soft prefactor for this assignment is $-|c|^2(q^2)^{-i\tilde{\nu}}(K^2)^{i\tilde{\nu}}$, from the opposite signs of the two branch prefactors displayed above. Its minus sign cancels the closed-loop sign in Eq. (32). The endpoint time powers are included when integrating these hard matrices. Relabelling $(\tau_3, \mathsf{a}_3) \leftrightarrow (\tau_4, \mathsf{a}_4)$ in the right matrix makes both integrated hard coefficients equal to $\mathcal{H}_1(-i\tilde{\nu}, +i\tilde{\nu}; 1)$. Each

hard end therefore has a Bose-even zeroth-order term. There are two soft assignments and two Higgs orderings at each end, giving $2 \times 2 \times 2$ contributions; the spin trace adds $2\widehat{\mathbf{q}} \cdot \widehat{\mathbf{K}}$. The soft coefficients give $|c|^2 = 1/(4\cosh^2 \pi\tilde{\nu})$. These factors fix the normalization of the mixed term independently of the oscillatory branch.

4.2 The mixed hard coefficient

The same Whittaker reduction, Eq. (84), gives

$$\mathcal{M}(\tilde{\nu}) = \lim_{\delta \to 0} \mathcal{H}_1(-\mathrm{i}\tilde{\nu} + \delta, +\mathrm{i}\tilde{\nu} + \delta; 1). \tag{105}$$

Its scalar seed powers start at $P = -3/2 - \mathrm{i}\tilde{\nu} + \delta$ and $Q = -3/2 + \mathrm{i}\tilde{\nu} + \delta$, with the integer shifts specified in Eq. (84). All SK branches and both $\kappa = \pm 1/2$ are combined before the regulator is removed. The exact conjugation relation $\mathcal{H}_1(\alpha, \beta; 1)^* = \mathcal{H}_1(\beta^*, \alpha^*; 1)$ makes $\mathcal{M}$ real. The implementation takes a fourth-order extrapolated common-regulator limit and checks the step size against the independent reference. For example, $\mathcal{M}(0.73) = -1.388500280198814$.

4.3 Central integral and total trispectrum

The scalar bubble in this case is

$$\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} (q^2)^{-\mathrm{i}\tilde{\nu}} (K^2)^{\mathrm{i}\tilde{\nu}} \widehat{\mathbf{q}} \cdot \widehat{\mathbf{K}} = \frac{k_s^3}{(4\pi)^{3/2}} \frac{2\sqrt{\pi}}{3} \tilde{\nu}(1 + \tilde{\nu}^2) \coth(\pi\tilde{\nu}). \tag{106}$$

After both soft branches and all four s-cut boxes are summed, the result is

$$\langle \delta h^4 \rangle' \big|_{s,\mathrm{mixed}} = \frac{N_c g_t^4 H^4 k_s^3}{16 \prod_i k_i^3} \frac{2\tilde{\nu}(1 + \tilde{\nu}^2)}{3\pi \sinh(2\pi\tilde{\nu})} \mathcal{M}(\tilde{\nu})^2. \tag{107}$$

It is independent of angles at this order. Equation (107) is the leading nonoscillatory nonanalytic term, whereas Eq. (97) is the leading mass-dependent clock signal. At $\tilde{\nu} = 0$ the separated branches coalesce; the limit must be taken after combining the relevant terms. Here $k_s^3 = (\mathbf{k}_s^2)^{3/2}$ is nonanalytic as a function of the soft momentum components, even though it has no mass-dependent oscillation. It cannot be discarded as a polynomial contact term. The remaining real coefficient is evaluated with the same scalar seed through Eq. (105). For example, $\mathcal{M}(0.73) = -1.388500280198814$.

The results above include the leading oscillatory and nonoscillatory nonanalytic terms for a single collapsed channel. Analytic contact terms, overlapping collapsed channels and Higgs-to-curvature conversion are outside the calculation. Section 5 specifies the numerical checks and their limits.

5 Numerical checks and scope

The numerical routines retain the parameter name `mu` for $\tilde{\nu} = m_t/H$ and output keys `B1`, `D1` for the angular coefficients B, D. Their definitions and normalization are unchanged. The Whittaker reduction is implemented in `checks/whittaker_scalar_seed.py` and tested by `checks/`

`whittaker_seed_checks.py`. The tests compare all three scalar-derived spinor propagators with independent Whittaker mode contractions, for every SK pair at $\tilde{\nu} = 0.2, 0.73, 2, 5$. They also check the two hard coefficients, the regulated hard integrals, the complete-SK limit and numerical precision stability. No auxiliary Hankel seed implementation is required. The checks below describe the additional propagator, contraction and collapsed-limit tests.

The Whittaker implementation constructs all normal and anomalous contractions directly from Eq. (11). It verifies EOMs, normalization and the Hankel map at $\tilde{\nu} = 0.2, 0.73, 2, 5$. The corresponding explicit Whittaker products are checked separately in `checks/explicit_propagator_checks.py`: all three propagators and all SK assignments agree with the mode contractions at four masses, three momentum directions and three time pairs. An alternative Hankel calculation uses explicit helicity spinors; the largest residual is 3.1×10^{-16}. Conjugation and equal-time anticommutators are checked as well. Raising special-function precision from 30 to 60 digits leaves the resulting double-precision matrices unchanged. These are free-propagator consistency checks, not an independent box integration. The independent eight-slot Wick enumerator gives 48 connected pairings and agrees with the cycle formula, and with half the Dirac result, for all 16 SK assignments at $\tilde{\nu} = 0.2, 0.73, 2$. Its test momenta are generic edge momenta: this is a local contraction test, not a momentum integral. The original-field implementation in `checks/aoki_weyl_checks.py` also compares both soft branches with an independent Bessel decomposition at $\tilde{\nu} = 0.2, 0.73, 2, 5$. It checks the four-type hard sums against Hankel scalar products, both mixed assignments, and the factorized spin cut for all 16 SK choices at the first three masses. The first-descendant remainder decreases quadratically with the soft momentum. These tests use the original $\psi, \psi^\dagger$ contractions. The soft-expansion check retains the untruncated nonlocal mode branches, Whittaker hard lines, both external energy shifts and all four orderings. On a finite time interval, at three masses and two geometries, its error relative to the leading nonzero coefficient decreases by approximately four when the soft momenta are halved. The nonzero single-cycle zeroth-order trace and its exchange cancellation are checked separately. Time-quadrature refinement is also tested.

A further independent calculation integrates the fermion Hankel kernels on BD-rotated contours at the physical mass indices: the two same-SK ordered triangles rotate together, and the two opposite-SK integrals rotate separately. It checks both $\mathcal{H}_1$ and $\mathcal{H}_0$ scalar hard integrals and the hard-momentum derivative at $\tilde{\nu} = 0.2, 0.73$, with a common positive endpoint regulator $\delta = 0.8$, including a mixed-branch case. The derivative is obtained directly from the first-order mode equations, independently of the scalar bootstrap source; the regulated $\mathcal{P}_1, \mathcal{P}_2$ are compared as well. This tests the scalar-seed normalization beyond the conformal-scalar example. It does not directly evaluate the zero-regulator limit; that limit still uses the analytically summed seeds and regulator-path checks.

The scalar-seed tests check regulator convergence and working precision. Comparing the regulated and analytically summed versions of the same Whittaker seed is a consistency check, not a second derivation of that seed. Neither finite-interval soft tests nor special-function agreement is a direct numerical integration of the full unexpanded BD box. The analytic contact part and its renormalization, overlapping collapsed channels, evolving masses, nonzero chemical potential and curvature conversion have not been computed. Reproducible commands and residuals are recorded in `checks/top_quark_audit.md`.

6 Top Quark Signal

6.1 Preliminaries

We next compute the Higgs four-point function generated by the top-quark loop. We use the external propagator (3) and the collapsed kinematics

$$\mathbf{k}_s = \mathbf{k}_1 + \mathbf{k}_2 = -(\mathbf{k}_3 + \mathbf{k}_4), \qquad k_s \ll k_i, \qquad k_{12} = k_1 + k_2, \qquad k_{34} = k_3 + k_4. \tag{108}$$

The Yukawa coupling and mass parameter are

$$g_t \equiv \frac{y_t}{\sqrt{2}}, \qquad m_t = g_t h_0, \qquad \tilde{\nu} \equiv \frac{m_t}{H} > 0, \qquad N_c = 3. \tag{109}$$

With $a = -1/(H\tau)$ and $\eta^{AB} = \mathrm{diag}(-1,1,1,1)$, we use

$$\{\gamma^A, \gamma^B\} = -2\eta^{AB}\mathbf{1}_4, \qquad \bar{T} = T^\dagger\gamma^0, \qquad \gamma^5 = \mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3, \qquad P_{L,R} = \frac{\mathbf{1}_4 \mp \gamma^5}{2}. \tag{110}$$

Here $e_A{}^\mu = a^{-1}\delta_A{}^\mu$, $(\gamma^0)^\dagger = \gamma^0$ and $(\gamma^i)^\dagger = -\gamma^i$. Let $t_{L,\mathrm{cov}}$ and $t_{R,\mathrm{cov}}$ denote the covariant left- and right-handed top fields, represented by four-component spinors satisfying $P_L t_{L,\mathrm{cov}} = t_{L,\mathrm{cov}}$ and $P_R t_{R,\mathrm{cov}} = t_{R,\mathrm{cov}}$. Their action is

$$\begin{aligned} S_t = \int \mathrm{d}^4x\sqrt{-g}\,\Big[&\mathrm{i}\bar{t}_{L,\mathrm{cov}}\gamma^A e_A{}^\mu \nabla_\mu t_{L,\mathrm{cov}} + \mathrm{i}\bar{t}_{R,\mathrm{cov}}\gamma^A e_A{}^\mu \nabla_\mu t_{R,\mathrm{cov}} \\ &- (m_t + g_t\delta h)(\bar{t}_{L,\mathrm{cov}} t_{R,\mathrm{cov}} + \bar{t}_{R,\mathrm{cov}} t_{L,\mathrm{cov}})\Big]. \end{aligned} \tag{111}$$

Color indices are suppressed. We combine the chiral fields into

$$T_{\mathrm{cov}} = t_{L,\mathrm{cov}} + t_{R,\mathrm{cov}}, \qquad t_{L,\mathrm{cov}} = P_L T_{\mathrm{cov}}, \qquad t_{R,\mathrm{cov}} = P_R T_{\mathrm{cov}}. \tag{112}$$

Using $\bar{t}_{L,\mathrm{cov}} = \bar{T}_{\mathrm{cov}} P_R$ and $\bar{t}_{R,\mathrm{cov}} = \bar{T}_{\mathrm{cov}} P_L$, the action becomes

$$S_t = \int \mathrm{d}^4x\sqrt{-g}\,\bar{T}_{\mathrm{cov}}\left(\mathrm{i}\gamma^A e_A{}^\mu \nabla_\mu - m_t - g_t\delta h\right) T_{\mathrm{cov}}. \tag{113}$$

Rescaling $T = a^{3/2} T_{\mathrm{cov}}$ removes the spin-connection term and gives

$$S_t = \int \mathrm{d}\tau\, \mathrm{d}^3\mathbf{x}\,\left[\bar{T}(\mathrm{i}\gamma^\mu\partial_\mu - a m_t)T - a g_t \delta h\, \bar{T}T\right], \qquad \gamma^\mu \equiv \delta_A{}^\mu \gamma^A. \tag{114}$$

We write $\gamma_k = \boldsymbol{\gamma}\cdot\widehat{\mathbf{k}}$, where $\boldsymbol{\gamma}\cdot\mathbf{v} = \sum_i \gamma^i v_i$. The helicity operator and projectors are $\Sigma_k = \gamma^5\gamma^0\gamma_k$ and $P_s = (\mathbf{1}_4 + s\Sigma_k)/2$, with $s = \pm 1$. Choose polarization spinors satisfying

$$P_L h^s_{\mathbf{k}} = h^s_{\mathbf{k}}, \qquad \Sigma_k h^s_{\mathbf{k}} = s h^s_{\mathbf{k}}, \qquad h^{s\dagger}_{\mathbf{k}} h^{s'}_{\mathbf{k}} = \delta^{ss'}, \qquad h^s_{\mathbf{k}} h^{s\dagger}_{\mathbf{k}} = P_L P_s. \tag{115}$$

The mode expansion is

$$T(\tau,\mathbf{x}) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \sum_{s=\pm1} \left[U_s(\mathbf{k},\tau) b^s_{\mathbf{k}} e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}} + V_s(\mathbf{k},\tau) d^{s\dagger}_{\mathbf{k}} e^{-\mathrm{i}\mathbf{k}\cdot\mathbf{x}}\right], \tag{116}$$

where $b^s_{\mathbf{k}}$ and $d^s_{\mathbf{k}}$ annihilate the top and antitop, with

$$\{b^s_{\mathbf{k}}, b^{s'\dagger}_{\mathbf{k}'}\} = \{d^s_{\mathbf{k}}, d^{s'\dagger}_{\mathbf{k}'}\} = (2\pi)^3\delta^{ss'}\delta^3(\mathbf{k}-\mathbf{k}'). \tag{117}$$

The positive- and negative-frequency spinors are

$$U_s = (u_s\mathbf{1}_4 + v_s\gamma^0)h^s_{\mathbf{k}}, \qquad V_s = C\bar{U}_s^T, \qquad \bar{U}_s = U_s^\dagger\gamma^0, \tag{118}$$

Here $C(\gamma^A)^TC^{-1} = -\gamma^A$, $C^T = -C$ and $C^\dagger C = \mathbf{1}_4$; the label s on V_s refers to U_s, so $\Sigma_k V_s = -sV_s$. The normalized Bunch–Davies mode functions are

$$u_+ = v_- = \frac{\tilde{\nu}}{\sqrt{2x}}W_{-1/2,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}x), \qquad u_- = v_+ = \frac{\mathrm{i}}{\sqrt{2x}}W_{1/2,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}x), \qquad x = -k\tau. \tag{119}$$

They satisfy

$$\mathrm{i}\partial_\tau u_s + sku_s = am_tv_s, \qquad \mathrm{i}\partial_\tau v_s - skv_s = am_tu_s, \qquad |u_s|^2 + |v_s|^2 = 1. \tag{120}$$

We use the propagator

$$D_{\mathsf{ab},\alpha\beta}(\mathbf{k};\tau_1,\tau_2) = \langle T_C T_{\mathsf{a},\alpha}(\mathbf{k},\tau_1)\bar{T}_{\mathsf{b},\beta}(-\mathbf{k},\tau_2)\rangle', \qquad \alpha,\beta = 1,\dots,4. \tag{121}$$

Here T_C denotes ordering along the Schwinger–Keldysh contour, and $\mathsf{a},\mathsf{b} = \pm 1$ are its branch labels. The Wightman components are

$$\begin{aligned} D_{-+}(\mathbf{k};\tau_1,\tau_2) &= \sum_s U_s(\mathbf{k},\tau_1)\bar{U}_s(\mathbf{k},\tau_2), \\ D_{+-}(\mathbf{k};\tau_1,\tau_2) &= -\sum_s V_s(-\mathbf{k},\tau_1)\bar{V}_s(-\mathbf{k},\tau_2). \end{aligned} \tag{122}$$

With $\theta_{ij} = \theta(\tau_i - \tau_j)$, the time-ordered components are

$$D_{++} = \theta_{12}D_{-+} + \theta_{21}D_{+-}, \qquad D_{--} = \theta_{21}D_{-+} + \theta_{12}D_{+-}. \tag{123}$$

The fermionic reversal rule is

$$\langle T_C\bar{T}_{\mathsf{a},\alpha}(\mathbf{k},\tau_1)T_{\mathsf{b},\beta}(-\mathbf{k},\tau_2)\rangle' = -[D_{\mathsf{ba}}(-\mathbf{k};\tau_2,\tau_1)]_{\beta\alpha}. \tag{124}$$

For the seed reduction, express the propagator through the scalar kernel with chemical potential [3]. With $X = -k\tau_1$ and $Y = -k\tau_2$, define

$$\begin{aligned} D^{(\kappa)}_{-+}(k;\tau_1,\tau_2) &= \frac{e^{\mathrm{i}\pi\kappa}}{2k}W_{\kappa,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}X)W_{-\kappa,\mathrm{i}\tilde{\nu}}(2\mathrm{i}Y), \\ D^{(\kappa)}_{+-}(k;\tau_1,\tau_2) &= \frac{e^{\mathrm{i}\pi\kappa}}{2k}W_{-\kappa,\mathrm{i}\tilde{\nu}}(2\mathrm{i}X)W_{\kappa,\mathrm{i}\tilde{\nu}}(-2\mathrm{i}Y). \end{aligned} \tag{125}$$

The time-ordered components obey (123). The seed parameter is analytically continued to $\kappa = \pm 1/2$. The Dirac propagator is then

$$\begin{aligned} D_{\mathsf{ab}} = \frac{k}{2\sqrt{XY}}\Big\{&\tilde{\nu}\left[D^{(-1/2)}_{\mathsf{ab}}(\mathbf{1}_4 - \gamma^0\gamma_k) + D^{(+1/2)}_{\mathsf{ab}}(\mathbf{1}_4 + \gamma^0\gamma_k)\right] \\ &+ \mathrm{i}\Big[\Big((-X\partial_X - \mathrm{i}X + \tfrac{1}{2})D^{(-1/2)}_{\mathsf{ab}}\Big)(\mathbf{1}_4 + \gamma^0\gamma_k) \\ &- \Big((X\partial_X - \mathrm{i}X - \tfrac{1}{2})D^{(+1/2)}_{\mathsf{ab}}\Big)(\mathbf{1}_4 - \gamma^0\gamma_k)\Big]\gamma^0\Big\}. \end{aligned} \tag{126}$$

The derivatives act on the scalar kernels at fixed Y. The soft limit follows from the small-argument expansion

$$W_{\kappa,\mathrm{i}\tilde{\nu}}(z) = \sum_{\eta=\pm1} \frac{\Gamma(-2\mathrm{i}\eta\tilde{\nu})}{\Gamma(\frac{1}{2} - \mathrm{i}\eta\tilde{\nu} - \kappa)} z^{1/2+\mathrm{i}\eta\tilde{\nu}} \left[1 - \frac{\kappa z}{1+2\mathrm{i}\eta\tilde{\nu}} + O(z^2)\right]. \tag{127}$$

Substituting this expansion gives, at $|k\tau_1|, |k\tau_2| \ll 1$, the required nonlocal terms

$$\begin{aligned}[D_{\mathsf{ab}}(\mathbf{k};\tau_1,\tau_2)]_{\mathrm{NonL}} = & -\frac{1}{4\pi} \sum_{\eta=\pm1} \Gamma^2(\tfrac{1}{2} - \mathrm{i}\eta\tilde{\nu}) \left(\frac{k^2\tau_1\tau_2}{4}\right)^{\mathrm{i}\eta\tilde{\nu}} \\ & \times \left[(\mathbf{1}_4 + \eta\gamma^0)\gamma_k + \frac{\mathrm{i}k}{1+2\mathrm{i}\eta\tilde{\nu}}\Big((\tau_1-\tau_2)\gamma^0 - \eta(\tau_1+\tau_2)\mathbf{1}_4\Big) + \cdots\right]. \end{aligned} \tag{128}$$

The omitted terms are second order in soft momentum at fixed times. These nonlocal terms are independent of the SK indices, allowing the two soft lines to be cut while the hard lines retain their full time dependence and SK ordering [2, 4]. The Yukawa interaction gives the amputated vertex $-\mathsf{i}\mathsf{a}g_t a(\tau)\mathbf{1}_4$.

6.2 The Box Diagram and Soft Expansion

We first consider the ordering 1234 in Fig. 2. Writing $D^{ij}(\mathbf{p}) = D_{\mathsf{a}_i\mathsf{a}_j}(\mathbf{p};\tau_i,\tau_j)$, the four-point function is

$$\begin{aligned}\langle \delta h^4\rangle'_{1234} = & -N_c g_t^4 \sum_{\mathsf{a}_1,\ldots,\mathsf{a}_4=\pm1} \left(\prod_{v=1}^4 \mathsf{a}_v\right) \int_{-\infty}^0 \prod_{v=1}^4 [\mathrm{d}\tau_v\, a(\tau_v) G_{\mathsf{a}_v}(k_v;\tau_v)] \\ & \times \int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \,\mathrm{tr}_4\Big[D^{12}(\mathbf{q}+\mathbf{k}_1)D^{23}(\mathbf{K})D^{34}(\mathbf{q}-\mathbf{k}_4)D^{41}(\mathbf{q})\Big], \qquad \mathbf{K} = \mathbf{q}+\mathbf{k}_s. \end{aligned} \tag{129}$$

The closed fermion loop supplies a minus sign and the color factor N_c. The 4! external attachments cancel the interaction-expansion factorial, leaving unit weight for each directed Dirac contraction. The six orderings are $1234, 1243, 1324, 1342, 1423, 1432$.

In the double-soft region $q, K \sim k_s$, introduce centered hard momenta

$$\mathbf{k}_L = \frac{\mathbf{k}_1 - \mathbf{k}_2}{2}, \qquad \mathbf{k}_R = \frac{\mathbf{k}_3 - \mathbf{k}_4}{2}, \qquad \boldsymbol{\ell} = \mathbf{q} + \frac{\mathbf{k}_s}{2}, \tag{130}$$

The hard lines carry $\mathbf{k}_L + \boldsymbol{\ell}$ and $\mathbf{k}_R + \boldsymbol{\ell}$. At leading collapsed order, $k_L \simeq k_{12}/2$ and $k_R \simeq k_{34}/2$. The cutting rule [4] replaces the two soft propagators by $D^{ij}_{\mathrm{NonL}} = [D^{ij}]_{\mathrm{NonL}}$. The four orderings contributing to this cut have spin factors

$$\begin{aligned}\mathcal{T}^{\mathrm{cut}}_{1234} &= \mathrm{tr}_4[D^{12}(\mathbf{k}_L+\boldsymbol{\ell})D^{23}_{\mathrm{NonL}}(\mathbf{K})D^{34}(\mathbf{k}_R+\boldsymbol{\ell})D^{41}_{\mathrm{NonL}}(\mathbf{q})], \\ \mathcal{T}^{\mathrm{cut}}_{1243} &= \mathrm{tr}_4[D^{12}(\mathbf{k}_L+\boldsymbol{\ell})D^{24}_{\mathrm{NonL}}(\mathbf{K})D^{43}(-\mathbf{k}_R+\boldsymbol{\ell})D^{31}_{\mathrm{NonL}}(\mathbf{q})], \\ \mathcal{T}^{\mathrm{cut}}_{1342} &= \mathrm{tr}_4[D^{21}(-\mathbf{k}_L+\boldsymbol{\ell})D^{13}_{\mathrm{NonL}}(\mathbf{K})D^{34}(\mathbf{k}_R+\boldsymbol{\ell})D^{42}_{\mathrm{NonL}}(\mathbf{q})], \\ \mathcal{T}^{\mathrm{cut}}_{1432} &= \mathrm{tr}_4[D^{21}(-\mathbf{k}_L+\boldsymbol{\ell})D^{14}_{\mathrm{NonL}}(\mathbf{K})D^{43}(-\mathbf{k}_R+\boldsymbol{\ell})D^{32}_{\mathrm{NonL}}(\mathbf{q})]. \end{aligned} \tag{131}$$

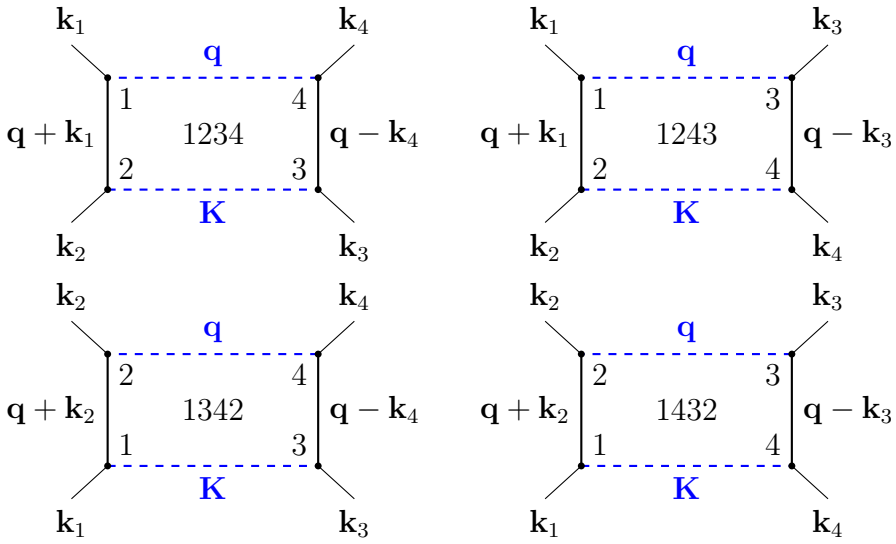


Figure 2: The four physical orderings in the collapsed s channel. Dashed lines carry the nonlocal soft propagators; solid internal lines are hard. Momentum labels are the arguments of the propagators in (131). Each vertex i retains $(\mathbf{k}_i, \tau_i, \mathsf{a}_i)$.

Every time and SK label remains attached to its physical external vertex, so all four terms have the same Higgs factors. The cut contribution is

$$[\langle \delta h^4 \rangle'_t]_{s,\mathrm{NonL}} \simeq -N_c g_t^4 \sum_{\mathsf{a}_1,\dots,\mathsf{a}_4=\pm 1} \Big(\prod_{v=1}^{4} \mathsf{a}_v\Big) \int_{-\infty}^{0} \prod_{v=1}^{4} [\mathrm{d}\tau_v\, a(\tau_v) G_{\mathsf{a}_v}(k_v;\tau_v)] \times \int_{\mathrm{soft}} \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \Big(\mathcal{T}^{\mathrm{cut}}_{1234} + \mathcal{T}^{\mathrm{cut}}_{1243} + \mathcal{T}^{\mathrm{cut}}_{1342} + \mathcal{T}^{\mathrm{cut}}_{1432}\Big). \tag{132}$$

Here $\int_{\mathrm{soft}}$ means the soft expansion at $q, K \sim k_s$, with the loop coefficients defined by analytic continuation.

We take the common branch $\eta = +1$ for example and suppress its label. Attaching the soft endpoints to each hard line gives matrices V_{12} and V_{34}, defined in App. A.1. With $\gamma_{\mathbf{n}} = \boldsymbol{\gamma}\cdot\mathbf{n}$ and $M(\mathbf{n}) = (\mathbf{1}_4 + \gamma^0)\gamma_{\mathbf{n}}$, the cut spin factor is

$$\mathcal{T}^{\mathrm{cut}}_{1234,+} = \frac{4^{-2\mathrm{i}\tilde{\nu}}\Gamma^4(\frac{1}{2} - \mathrm{i}\tilde{\nu})}{(4\pi)^2} (qK)^{2\mathrm{i}\tilde{\nu}} \prod_{v=1}^{4} (-\tau_v)^{\mathrm{i}\tilde{\nu}} \times \mathrm{tr}_4[V_{12} M(\widehat{\mathbf{K}}) V_{34} M(\widehat{\mathbf{q}})]. \tag{133}$$

The four orderings replace the trace by

$$\mathrm{tr}_4[(V_{12} + V_{21}) M(\widehat{\mathbf{K}}) (V_{34} + V_{43}) M(\widehat{\mathbf{q}})]. \tag{134}$$

To expand the hard ends, define

$$\Lambda_\pm = \frac{\mathbf{1}_4 \pm \gamma^0}{2}, \qquad M(\mathbf{n}) = 2\Lambda_+ \gamma_{\mathbf{n}} = 2\gamma_{\mathbf{n}} \Lambda_-. \tag{135}$$

Only $\Lambda_- V \Lambda_+$ enters the trace. At zeroth soft order, writing the scalar coefficient as $F_{0;\mathsf{ab}}$ gives

$$\begin{aligned}\Lambda_- V_{12}^{(0)} \Lambda_+ &= \frac{1}{2} F_{0;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\Lambda_-\gamma_{\mathbf{n}_L}\Lambda_+,\\ \Lambda_- V_{21}^{(0)} \Lambda_+ &= -\frac{1}{2} F_{0;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\Lambda_-\gamma_{\mathbf{n}_L}\Lambda_+, \qquad \mathbf{n}_L = \mathbf{k}_L/k_L. \end{aligned} \tag{136}$$

The right pair obeys the same relation with $(L,1,2) \to (R,3,4)$. The coefficient $F_{0;\mathsf{ab}}$ is invariant under reversal of its time and SK labels. Thus each projected ordering sum vanishes at the integrand level, removing the candidate $k_s^{3\pm 4i\tilde{\nu}}$ signal. Every term with a zeroth-order hard end vanishes, so the first surviving term is first order on both sides.

The first-order corrections to the hard and soft propagators give

$$\begin{aligned}\Lambda_-(V_{12}^{(1)} + V_{21}^{(1)})\Lambda_+ &= \frac{1}{k_L}\Lambda_-(\boldsymbol{\gamma}\cdot\mathbf{v}_L)\Lambda_+, \qquad \mathbf{v}_L = \Pi_1^L \boldsymbol{\ell} + \Pi_2^L \mathbf{n}_L(\mathbf{n}_L\cdot\boldsymbol{\ell}),\\ \Lambda_-(V_{34}^{(1)} + V_{43}^{(1)})\Lambda_+ &= \frac{1}{k_R}\Lambda_-(\boldsymbol{\gamma}\cdot\mathbf{v}_R)\Lambda_+, \qquad \mathbf{v}_R = \Pi_1^R \boldsymbol{\ell} + \Pi_2^R \mathbf{n}_R(\mathbf{n}_R\cdot\boldsymbol{\ell}).\end{aligned} \tag{137}$$

Here $\mathbf{n}_R = \widehat{\mathbf{k}}_R$ and $\Pi_i^L = \Pi_{i;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)$, with the analogous right definition. The scalar functions Π_i are given in (157). Using

$$\begin{aligned}&\mathrm{tr}_4[\Lambda_-(\boldsymbol{\gamma}\cdot\mathbf{a})(\boldsymbol{\gamma}\cdot\mathbf{b})(\boldsymbol{\gamma}\cdot\mathbf{c})(\boldsymbol{\gamma}\cdot\mathbf{d})]\\ &\quad = 2[(\mathbf{a}\cdot\mathbf{b})(\mathbf{c}\cdot\mathbf{d}) - (\mathbf{a}\cdot\mathbf{c})(\mathbf{b}\cdot\mathbf{d}) + (\mathbf{a}\cdot\mathbf{d})(\mathbf{b}\cdot\mathbf{c})],\end{aligned} \tag{138}$$

the leading term of the sum of four traces becomes

$$\begin{aligned}&\mathrm{tr}_4[(V_{12}^{(1)} + V_{21}^{(1)})M(\widehat{\mathbf{K}})(V_{34}^{(1)} + V_{43}^{(1)})M(\widehat{\mathbf{q}})]\\ &= \frac{8}{k_L k_R q K}\left[2(\mathbf{v}_L\cdot\boldsymbol{\ell})(\mathbf{v}_R\cdot\boldsymbol{\ell}) - \frac{1}{2}(\mathbf{v}_L\cdot\mathbf{k}_s)(\mathbf{v}_R\cdot\mathbf{k}_s) - (\mathbf{v}_L\cdot\mathbf{v}_R)(\ell^2 - k_s^2/4)\right].\end{aligned} \tag{139}$$

The factor 8 consists of 2 from the projected Dirac trace and 4 from the two central matrices. Consequently the first nonzero oscillatory signal scales as $k_s^{5\pm 4i\tilde{\nu}}$.

6.3 Factorized Loop and Time Integrals

The scalar trace in (139) reduces the remaining loop to a scalar integral. We use the angular convention

$$\widehat{\mathbf{k}}_s = (0,0,1), \qquad \mathbf{n}_L = (\sin\theta_1, 0, \cos\theta_1), \qquad \mathbf{n}_R = (\sin\theta_3\cos\phi_3, \sin\theta_3\sin\phi_3, \cos\theta_3). \tag{140}$$

For each common soft branch $\eta = \pm 1$, put $\mathbf{v}_{A,\eta} = \Pi_{1,\eta}^A \boldsymbol{\ell} + \Pi_{2,\eta}^A \mathbf{n}_A(\mathbf{n}_A\cdot\boldsymbol{\ell})$, with $A = L, R$. The dimensionless loop coefficients are defined by

$$\begin{aligned}&\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}(qK)^{2i\eta\tilde{\nu}-1}\left[2(\mathbf{v}_{L,\eta}\cdot\boldsymbol{\ell})(\mathbf{v}_{R,\eta}\cdot\boldsymbol{\ell}) - \frac{1}{2}(\mathbf{v}_{L,\eta}\cdot\mathbf{k}_s)(\mathbf{v}_{R,\eta}\cdot\mathbf{k}_s) - (\mathbf{v}_{L,\eta}\cdot\mathbf{v}_{R,\eta})(\ell^2 - k_s^2/4)\right]\\ &\quad = \frac{k_s^{5+4i\eta\tilde{\nu}}}{(4\pi)^{3/2}}\sum_{i,j=1}^{2}\mathcal{I}_t^{ij}(-i\eta\tilde{\nu},\theta_1,\theta_3,\phi_3)\Pi_{i,\eta}^L\Pi_{j,\eta}^R.\end{aligned} \tag{141}$$

The resulting coefficients satisfy

$$\mathcal{I}_t^{11} = 3\mathcal{I}_t^{12} = 3\mathcal{I}_t^{21}. \tag{142}$$

These three coefficients are independent of angles. All angular dependence is carried by $\mathcal{I}_t^{22}$, whose explicit expression is given in App. A.1. For one common soft branch, the loop summed over the four orderings is

$$\begin{aligned}\int_{\text{soft}} \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \sum_{\text{ord}} \mathcal{T}^{\text{cut}}_{\text{ord},\eta} \simeq \frac{8\, 4^{-2\mathrm{i}\eta\tilde{\nu}}\Gamma^4(\frac{1}{2} - \mathrm{i}\eta\tilde{\nu}) k_s^{5+4\mathrm{i}\eta\tilde{\nu}}}{(4\pi)^{7/2} k_L k_R} \prod_{v=1}^{4} (-\tau_v)^{\mathrm{i}\eta\tilde{\nu}} \\ \times \sum_{i,j=1}^{2} \mathcal{I}_t^{ij}(-\mathrm{i}\eta\tilde{\nu}, \theta_1, \theta_3, \phi_3) \Pi^L_{i,\eta} \Pi^R_{j,\eta}.\end{aligned} \tag{143}$$

The two time integrations on each side are now independent. For the left subgraph, the leading collapsed term at $k_1 \simeq k_2 \simeq k_L \simeq k_{12}/2$ defines

$$\begin{aligned}&\sum_{\mathsf{a},\mathsf{b}=\pm 1} \mathsf{ab} \int_{-\infty}^{0} \frac{\mathrm{d}\tau_1}{H\tau_1} \frac{\mathrm{d}\tau_2}{H\tau_2} (-\tau_1)^{\mathrm{i}\eta\tilde{\nu}} (-\tau_2)^{\mathrm{i}\eta\tilde{\nu}} G_{\mathsf{a}}(k_1;\tau_1) G_{\mathsf{b}}(k_2;\tau_2) \Pi_{i,\eta;\mathsf{ab}}(k_{12}/2;\tau_1,\tau_2) \\ &\quad \simeq \frac{H^2}{4k_1^3 k_2^3} k_{12}^{-2\mathrm{i}\eta\tilde{\nu}} \mathcal{P}_{t,i,\eta}(\tilde{\nu}), \qquad i = 1, 2.\end{aligned} \tag{144}$$

Below, $\mathcal{P}_{t,i} \equiv \mathcal{P}_{t,i,+}$ denotes the positive-branch coefficient. The right subgraph follows by $(1,2,L) \to (3,4,R)$ and $k_{12} \to k_{34}$. The seed normalization, its closed form and the combinations giving $\mathcal{P}_{t,i}$ are given in App. A.2.

6.4 Oscillatory Signal

Combining the soft loop and the two hard subgraphs, setting $N_c = 3$, and adding the complex-conjugate branch gives

$$\left[\langle \delta h^4 \rangle'_t\right]_{\text{NS}} \simeq -\frac{3 g_t^4 H^4 k_s^5}{4k_1^3 k_2^3 k_3^3 k_4^3 k_{12} k_{34}} \left[\left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2\mathrm{i}\tilde{\nu}} C_t(\tilde{\nu}, \theta_1, \theta_3, \phi_3) + \text{c.c.} \right], \tag{145}$$

where

$$C_t = \frac{8\Gamma^4(\frac{1}{2} - \mathrm{i}\tilde{\nu})}{(4\pi)^{7/2}} \sum_{i,j=1}^{2} \mathcal{P}_{t,i} \mathcal{P}_{t,j} \mathcal{I}_t^{ij}(-\mathrm{i}\tilde{\nu}, \theta_1, \theta_3, \phi_3). \tag{146}$$

The angular decomposition is

$$\begin{aligned}C_t = A_{1,t} + A_{2,t}(\cos^2\theta_1 + \cos^2\theta_3) + A_{3,t} \cos^2\theta_1 \cos^2\theta_3 \\ + B_t \sin(2\theta_1)\sin(2\theta_3)\cos\phi_3 + D_t \sin^2\theta_1 \sin^2\theta_3 \cos(2\phi_3).\end{aligned} \tag{147}$$

The five coefficients depend only on $\tilde{\nu}$ and satisfy $A_{3,t} = -3A_{2,t}$. Their analytic expressions are given in App. A.3. All nonconstant angular terms are proportional to $\mathcal{P}_{t,2}^2$. At fixed angles the oscillatory bracket in (145) is

$$2|C_t| \cos\left[2\tilde{\nu} \log\left(\frac{k_s^2}{4k_{12}k_{34}} \right) + \arg C_t \right]. \tag{148}$$

This is the leading same-branch oscillatory contribution in the collapsed s channel. The other channels follow by external permutations.

A Top Quark Integrals and Coefficients

A.1 Soft Expansion and Loop Integrals

For either soft branch $\eta = \pm 1$, a nonlocal line factorizes as

$$D^{ij}_{\mathrm{NonL},\eta}(\mathbf{p}) = -\frac{\Gamma^2(\frac{1}{2} - \mathrm{i}\eta\tilde{\nu})}{4\pi}\left(\frac{p^2\tau_i\tau_j}{4}\right)^{\mathrm{i}\eta\tilde{\nu}} L_\eta(\mathbf{p},\tau_i) M_\eta(\widehat{\mathbf{p}}) U_\eta(\mathbf{p},\tau_j),$$
$$M_\eta(\widehat{\mathbf{p}}) = (\mathbf{1}_4 + \eta\gamma^0)\gamma_p, \qquad L_\eta(\mathbf{p},\tau) = U_\eta(\mathbf{p},\tau) = \mathbf{1}_4 + \frac{\mathrm{i}\eta p\tau}{1 + 2\mathrm{i}\eta\tilde{\nu}}\gamma_p + O(p^2\tau^2). \tag{149}$$

The unexpanded endpoint functions are

$$L_\eta(\mathbf{p},\tau) = U_\eta(\mathbf{p},\tau) = {}_0F_1\left(;\tfrac{1}{2} + \mathrm{i}\eta\tilde{\nu}; -\frac{p^2\tau^2}{4}\right)\mathbf{1}_4 + \frac{\mathrm{i}\eta p\tau}{1 + 2\mathrm{i}\eta\tilde{\nu}}{}_0F_1\left(;\tfrac{3}{2} + \mathrm{i}\eta\tilde{\nu}; -\frac{p^2\tau^2}{4}\right)\gamma_p. \tag{150}$$

The hard matrices are

$$\begin{aligned} V_{12,\eta} &= U_\eta(\mathbf{q},\tau_1) D^{12}(\mathbf{k}_L + \boldsymbol{\ell}) L_\eta(\mathbf{K},\tau_2), \\ V_{21,\eta} &= U_\eta(\mathbf{q},\tau_2) D^{21}(-\mathbf{k}_L + \boldsymbol{\ell}) L_\eta(\mathbf{K},\tau_1), \\ V_{34,\eta} &= U_\eta(\mathbf{K},\tau_3) D^{34}(\mathbf{k}_R + \boldsymbol{\ell}) L_\eta(\mathbf{q},\tau_4), \\ V_{43,\eta} &= U_\eta(\mathbf{K},\tau_4) D^{43}(-\mathbf{k}_R + \boldsymbol{\ell}) L_\eta(\mathbf{q},\tau_3). \end{aligned} \tag{151}$$

We write $V_{ij} = V_{ij,+}$ in the positive-branch derivation. For $\Lambda_\eta = (\mathbf{1}_4 + \eta\gamma^0)/2$, define two scalar functions to express the subgraph contributions:

$$\begin{aligned} F_{0,\eta;\mathsf{ab}}(p;\tau_1,\tau_2) &= -\operatorname{tr}_4[\Lambda_{-\eta} D_{\mathsf{ab}}(p\mathbf{n};\tau_1,\tau_2)\gamma_{\mathbf{n}}], \\ F_{1,\eta;\mathsf{ab}}(p;\tau_1,\tau_2) &= -\eta \operatorname{tr}_4[\Lambda_\eta D_{\mathsf{ab}}(p\mathbf{n};\tau_1,\tau_2)]. \end{aligned} \tag{152}$$

With $X = -p\tau_1$, $Y = -p\tau_2$, the scalar reconstruction (126) gives

$$\begin{aligned} F_{0,\eta;\mathsf{ab}}(p;\tau_1,\tau_2) &= -\frac{\mathrm{i}p}{\sqrt{XY}}\left[(X\partial_X - \mathrm{i}X - \tfrac{1}{2} - \mathrm{i}\eta\tilde{\nu})D^{(+1/2)}_{\mathsf{ab}} + (-X\partial_X - \mathrm{i}X + \tfrac{1}{2} + \mathrm{i}\eta\tilde{\nu})D^{(-1/2)}_{\mathsf{ab}}\right], \\ F_{1,\eta;\mathsf{ab}}(p;\tau_1,\tau_2) &= -\frac{\mathrm{i}p}{\sqrt{XY}}\left[(-X\partial_X - \mathrm{i}X + \tfrac{1}{2} - \mathrm{i}\eta\tilde{\nu})D^{(-1/2)}_{\mathsf{ab}} - (X\partial_X - \mathrm{i}X - \tfrac{1}{2} + \mathrm{i}\eta\tilde{\nu})D^{(+1/2)}_{\mathsf{ab}}\right]. \end{aligned} \tag{153}$$

We write $F_j = F_{j,+}$. Since $F_{0,\eta;\mathsf{ab}}(p;\tau_1,\tau_2) = F_{0,\eta;\mathsf{ba}}(p;\tau_2,\tau_1)$, reversing the hard momentum changes the sign of $\Lambda_{-\eta} D \Lambda_\eta$, proving (136). The projections needed through first order are

$$\begin{aligned} \Lambda_{-\eta} D_{\mathsf{ab}} \Lambda_\eta &= \tfrac{1}{2} F_{0,\eta;\mathsf{ab}} \Lambda_{-\eta}\gamma_{\mathbf{n}}\Lambda_\eta, \\ \Lambda_\eta D_{\mathsf{ab}}\Lambda_\eta &= -\tfrac{\eta}{2} F_{1,\eta;\mathsf{ab}}\Lambda_\eta, \qquad \Lambda_{-\eta} D_{\mathsf{ab}}\Lambda_{-\eta} = -\tfrac{\eta}{2} F_{1,\eta;\mathsf{ba}}(p;\tau_2,\tau_1)\Lambda_{-\eta}. \end{aligned} \tag{154}$$

The hard-line correction in $V_{12,\eta}$ is

$$\Lambda_{-\eta} V^{(1)}_{12,\eta}\Lambda_\eta\big|_{\mathrm{hard}} = \frac{1}{2k_L}\Lambda_{-\eta}\boldsymbol{\gamma}\cdot\left[F_{0,\eta;\mathsf{a}_1\mathsf{a}_2}\boldsymbol{\ell} + (p\partial_p - 1)F_{0,\eta;\mathsf{a}_1\mathsf{a}_2}\mathbf{n}_L(\mathbf{n}_L\cdot\boldsymbol{\ell})\right]_{p=k_L}\Lambda_\eta. \tag{155}$$

while the two soft endpoints supply

$$\Lambda_{-\eta} V^{(1)}_{12,\eta} \Lambda_\eta \Big|_{\rm end} = -\frac{\mathrm{i}}{2(1+2\mathrm{i}\eta\tilde\nu)} \Lambda_{-\eta} \boldsymbol{\gamma}\cdot[\tau_1 F_{1,\eta;\mathsf{a}_1\mathsf{a}_2}(k_L;\tau_1,\tau_2)\mathbf{q} + \tau_2 F_{1,\eta;\mathsf{a}_2\mathsf{a}_1}(k_L;\tau_2,\tau_1)\mathbf{K}]\,\Lambda_\eta. \quad (156)$$

The reversed ordering has the same hard correction and interchanges the coefficients of $\mathbf{q}$ and $\mathbf{K}$. Adding the two, with $\mathbf{q}+\mathbf{K}=2\boldsymbol{\ell}$, gives (137), where

$$\begin{aligned}
\Pi_{1,\eta;\mathsf{ab}}(k;\tau_1,\tau_2) &= F_{0,\eta;\mathsf{ab}}(k;\tau_1,\tau_2) - \frac{\mathrm{i}k}{1+2\mathrm{i}\eta\tilde\nu}[\tau_1 F_{1,\eta;\mathsf{ab}}(k;\tau_1,\tau_2) + \tau_2 F_{1,\eta;\mathsf{ba}}(k;\tau_2,\tau_1)],\\
\Pi_{2,\eta;\mathsf{ab}}(k;\tau_1,\tau_2) &= (p\partial_p - 1) F_{0,\eta;\mathsf{ab}}(p;\tau_1,\tau_2)\big|_{p=k}\,.
\end{aligned} \quad (157)$$

The momentum derivative is at fixed times, and $\Pi_{i;\mathsf{ab}} \equiv \Pi_{i,+;\mathsf{ab}}$. For real $\tilde\nu$, $F_{0,-;\mathsf{ab}} = F^*_{0,+;-\mathsf{a},-\mathsf{b}}$ and $F_{1,-;\mathsf{ab}} = -F^*_{1,+;-\mathsf{a},-\mathsf{b}}$, implying $\mathcal{P}_{t,i,-} = \mathcal{P}^*_{t,i,+}$.

The loop moments reduce to the scalar convolution

$$\begin{aligned}
\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \frac{1}{(q^2)^\alpha (K^2)^\beta} &= \frac{k_s^{3-2\alpha-2\beta}}{(4\pi)^{3/2}} f_0(\alpha,\beta),\\
f_0(\alpha,\beta) &= \frac{\Gamma(\alpha+\beta-3/2)\Gamma(3/2-\alpha)\Gamma(3/2-\beta)}{\Gamma(3-\alpha-\beta)\Gamma(\alpha)\Gamma(\beta)}.
\end{aligned} \quad (158)$$

With $z = \widehat{\mathbf{q}}\cdot\widehat{\mathbf{k}}_s$ and $x = -\mathrm{i}\eta\tilde\nu$, define

$$\begin{aligned}
\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} q^{m-2x} K^{-n-2x} z^r &= \frac{k_s^{3+m-n-4x}}{(4\pi)^{3/2}} \mathcal{J}_r(m,n;x),\\
\mathcal{J}_r(m,n;x) &= \frac{1}{2^r} \sum_{u+v+w=r} \frac{r!}{u!v!w!} (-1)^{v+w} f_0\Big(x - \frac{m}{2} + \frac{r}{2} - v, x + \frac{n}{2} - u\Big).
\end{aligned} \quad (159)$$

The sum runs over nonnegative integers and follows from $z = (K^2 - q^2 - k_s^2)/(2qk_s)$. The convolutions are defined by analytic continuation, retaining the nonanalytic term.

For the loop coefficients, set $x = -\mathrm{i}\tilde\nu$. We use $\mathcal{J}_r(m,n;x)$ as defined in (159). The common argument x of $\mathcal{I}^{ij}_t$ and $\mathcal{J}_r$, and the angular arguments of $\mathcal{I}^{ij}_t$, are suppressed below. Azimuthal averaging and the scalar-convolution identities give

$$\begin{aligned}
\mathcal{I}^{11}_t &= 3\mathcal{I}^{12}_t = 3\mathcal{I}^{21}_t,\\
\mathcal{I}^{12}_t &= \frac{1}{2}[\mathcal{J}_0(3,1) - \mathcal{J}_2(3,1) + \mathcal{J}_1(2,1) - \mathcal{J}_3(2,1)] + \frac{1}{4}[\mathcal{J}_0(1,1) - \mathcal{J}_2(1,1)],\\
\mathcal{I}^{22}_t &= \frac{1}{4}[2\mathcal{J}_4(3,1) - 3\mathcal{J}_2(3,1) + \mathcal{J}_0(3,1) + \mathcal{J}_3(2,1) - \mathcal{J}_1(2,1)]\\
&\quad + \frac{1}{4}\Big[-6\mathcal{J}_4(3,1) + 7\mathcal{J}_2(3,1) - \mathcal{J}_0(3,1) + 5\mathcal{J}_1(2,1) - 5\mathcal{J}_3(2,1)\\
&\qquad + \mathcal{J}_0(1,1) - \mathcal{J}_2(1,1)\Big](\cos^2\theta_1 + \cos^2\theta_3 - 3\cos^2\theta_1\cos^2\theta_3)\\
&\quad + \frac{1}{16}\Big[-16\mathcal{J}_4(3,1) + 14\mathcal{J}_2(3,1) - 2\mathcal{J}_0(3,1) + 10\mathcal{J}_1(2,1) - 18\mathcal{J}_3(2,1)\\
&\qquad + 2\mathcal{J}_0(1,1) - 7\mathcal{J}_2(1,1) - \mathcal{J}_1(0,1)\Big]\sin(2\theta_1)\sin(2\theta_3)\cos\phi_3\\
&\quad + \frac{1}{4}[\mathcal{J}_4(3,1) - \mathcal{J}_2(3,1) + \mathcal{J}_3(2,1) - \mathcal{J}_1(2,1)]\sin^2\theta_1\sin^2\theta_3\cos(2\phi_3).
\end{aligned} \quad (160)$$

The finite sum (159) and the Gamma recurrence give the closed form below, with the common factor

$$\mathcal{G}_t(\tilde{\nu}) = \frac{\Gamma(-\frac{1}{2} - 2\mathrm{i}\tilde{\nu})\Gamma^2(1+\mathrm{i}\tilde{\nu})}{\Gamma(2+2\mathrm{i}\tilde{\nu})\Gamma^2(\frac{1}{2}-\mathrm{i}\tilde{\nu})}. \tag{161}$$

For $x = -\mathrm{i}\tilde{\nu}$, the fully reduced coefficients are

$$\begin{aligned}
\mathcal{I}_t^{12}(-\mathrm{i}\tilde{\nu}) &= -\mathcal{G}_t \frac{(1+\mathrm{i}\tilde{\nu})(1+2\mathrm{i}\tilde{\nu})}{4(3+2\mathrm{i}\tilde{\nu})(5+4\mathrm{i}\tilde{\nu})(3+4\mathrm{i}\tilde{\nu})}, \\
\mathcal{I}_t^{22}(-\mathrm{i}\tilde{\nu},\theta_1,\theta_3,\phi_3) &= -\frac{\mathcal{G}_t(1+\mathrm{i}\tilde{\nu})}{4(5+2\mathrm{i}\tilde{\nu})(3+2\mathrm{i}\tilde{\nu})(5+4\mathrm{i}\tilde{\nu})(3+4\mathrm{i}\tilde{\nu})} \\
&\quad\times\Big\{(2+\mathrm{i}\tilde{\nu})(1+2\mathrm{i}\tilde{\nu}) \\
&\quad-(1+\mathrm{i}\tilde{\nu})(1+2\mathrm{i}\tilde{\nu})[\cos^2\theta_1+\cos^2\theta_3-3\cos^2\theta_1\cos^2\theta_3] \\
&\quad-\frac{1}{2}(1+2\mathrm{i}\tilde{\nu})(3+2\mathrm{i}\tilde{\nu})\sin(2\theta_1)\sin(2\theta_3)\cos\phi_3 \\
&\quad+(2+\mathrm{i}\tilde{\nu})(3+2\mathrm{i}\tilde{\nu})\sin^2\theta_1\sin^2\theta_3\cos(2\phi_3)\Big\}.
\end{aligned} \tag{162}$$

The $\eta = -1$ branch follows by complex conjugation.

A.2 Whittaker Seed and Time Integrals

We denote the Whittaker seed $\mathcal{I}$ of Ref. [3] by $\mathcal{K}$. For equal external energies k, internal magnitude $p = rk$, and $u = 2r/(1+r)$, the integral seed is defined by

$$\mathcal{K}_{\mathsf{ab}}^{\kappa;p_1,p_2}(u,u) \equiv (-\mathsf{ab})p^{3+p_1+p_2}\int_{-\infty}^{0}\mathrm{d}\tau_1\mathrm{d}\tau_2\,(-\tau_1)^{p_1}(-\tau_2)^{p_2}e^{\mathrm{i}\mathsf{a}k\tau_1+\mathrm{i}\mathsf{b}k\tau_2}D_{\mathsf{ab}}^{(\kappa)}(p;\tau_1,\tau_2). \tag{163}$$

In terms of $x_1 = -k\tau_1$, $x_2 = -k\tau_2$, its time moment is

$$\int_0^\infty \mathrm{d}x_1\mathrm{d}x_2\,x_1^{p_1}x_2^{p_2}e^{-\mathrm{i}\mathsf{a}x_1-\mathrm{i}\mathsf{b}x_2}kD_{\mathsf{ab}}^{(\kappa)}(rk;-x_1/k,-x_2/k) = -\mathsf{ab}\,r^{-3-p_1-p_2}\mathcal{K}_{\mathsf{ab}}^{\kappa;p_1,p_2}(u,u). \tag{164}$$

At $r=1$, write $\mathcal{K}_{\mathsf{ab}}^{\kappa;p_1,p_2} \equiv \mathcal{K}_{\mathsf{ab}}^{\kappa;p_1,p_2}(1,1)$. We use $\Gamma[a_1,\dots,a_n] = \prod_j\Gamma(a_j)$. A tilde on ${}_3F_2$ divides the ordinary hypergeometric function by $\Gamma(b_1)\Gamma(b_2)$, where b_1,b_2 are its lower parameters. The closed expressions are

$$\mathcal{K}_{\pm\mp}^{\kappa;p_1,p_2} = \frac{e^{\mp\mathrm{i}\pi(p_1-p_2)/2+\mathrm{i}\pi\kappa}}{2^{3+p_1+p_2}}\frac{\Gamma[\frac{3}{2}+p_1-\mathrm{i}\tilde{\nu},\frac{3}{2}+p_1+\mathrm{i}\tilde{\nu},\frac{3}{2}+p_2-\mathrm{i}\tilde{\nu},\frac{3}{2}+p_2+\mathrm{i}\tilde{\nu}]}{\Gamma[2+p_1\pm\kappa,2+p_2\mp\kappa]}, \tag{165}$$

$$\begin{aligned}
\mathcal{K}_{\pm\pm}^{\kappa;p_1,p_2} &= \frac{\mp\mathrm{i}e^{\mp\mathrm{i}\pi(p_1+p_2)/2}}{2^{3+p_1+p_2}}\Gamma[\tfrac{3}{2}+p_1\pm\mathrm{i}\tilde{\nu},\tfrac{3}{2}+p_2\pm\mathrm{i}\tilde{\nu}] \\
&\quad\times\Bigg\{\frac{e^{2\mathrm{i}\pi\kappa}+e^{-2\pi\tilde{\nu}}}{2\pi}\frac{\Gamma[\frac{3}{2}+p_1\mp\mathrm{i}\tilde{\nu},\frac{3}{2}+p_2\mp\mathrm{i}\tilde{\nu},\frac{1}{2}\pm\kappa-\mathrm{i}\tilde{\nu},\frac{1}{2}\pm\kappa+\mathrm{i}\tilde{\nu}]}{\Gamma[2+p_1\pm\kappa,2+p_2\pm\kappa]} \\
&\quad\pm\mathrm{i}\Gamma(3+p_1+p_2){}_3\widetilde{F}_2\left[\begin{matrix}3+p_1+p_2,\ \frac{1}{2}\mp\kappa\pm\mathrm{i}\tilde{\nu},\ 1\\ \frac{5}{2}+p_1\pm\mathrm{i}\tilde{\nu},\ \frac{5}{2}+p_2\pm\mathrm{i}\tilde{\nu}\end{matrix}\middle|1\right]\Bigg\}.
\end{aligned} \tag{166}$$

Only two scalar time moments are needed to express all these hard coefficients. Define

$$\begin{aligned}
\mathcal{H}_j(q_1,q_2;r) &= \sum_{\mathsf{a},\mathsf{b}=\pm1}\mathsf{ab}\int_0^\infty \mathrm{d}x_1\mathrm{d}x_2\,x_1^{q_1-1}x_2^{q_2-1}(1+\mathrm{i}\mathsf{a}x_1)(1+\mathrm{i}\mathsf{b}x_2)e^{-\mathrm{i}\mathsf{a}x_1-\mathrm{i}\mathsf{b}x_2} \\
&\quad\times F_{j;\mathsf{ab}}(rk;-x_1/k,-x_2/k), \qquad j=0,1.
\end{aligned} \tag{167}$$

Scale invariance makes the right-hand side independent of k. Expanding the external polynomials with indices $m, n = 0, 1$, put $p_1 = q_1 - 3/2 + m$, $p_2 = q_2 - 3/2 + n$. Integration by parts sends $x_1\partial_{x_1}$ to $-(p_1+1) + \mathsf{i}\mathsf{a}x_1$. Using the definition (163) therefore gives directly

$$\mathcal{H}_0(q_1,q_2;r) = \mathrm{i}\sum_{\mathsf{a},\mathsf{b}=\pm1}\sum_{m,n=0}^{1}(\mathsf{i}\mathsf{a})^m(\mathsf{i}\mathsf{b})^n r^{-3-p_1-p_2}\Big\{(p_1+\tfrac{3}{2}+\mathrm{i}\tilde{\nu})[\mathcal{K}_{\mathsf{a}\mathsf{b}}^{-;p_1,p_2}-\mathcal{K}_{\mathsf{a}\mathsf{b}}^{+;p_1,p_2}] + \frac{\mathrm{i}(\mathsf{a}-r)}{r}\mathcal{K}_{\mathsf{a}\mathsf{b}}^{+;p_1+1,p_2} - \frac{\mathrm{i}(\mathsf{a}+r)}{r}\mathcal{K}_{\mathsf{a}\mathsf{b}}^{-;p_1+1,p_2}\Big\}, \tag{168}$$

$$\mathcal{H}_1(q_1,q_2;r) = \mathrm{i}\sum_{\mathsf{a},\mathsf{b}=\pm1}\sum_{m,n=0}^{1}(\mathsf{i}\mathsf{a})^m(\mathsf{i}\mathsf{b})^n r^{-3-p_1-p_2}\Big\{(p_1+\tfrac{3}{2}-\mathrm{i}\tilde{\nu})[\mathcal{K}_{\mathsf{a}\mathsf{b}}^{-;p_1,p_2}+\mathcal{K}_{\mathsf{a}\mathsf{b}}^{+;p_1,p_2}] - \frac{\mathrm{i}(\mathsf{a}-r)}{r}\mathcal{K}_{\mathsf{a}\mathsf{b}}^{+;p_1+1,p_2} - \frac{\mathrm{i}(\mathsf{a}+r)}{r}\mathcal{K}_{\mathsf{a}\mathsf{b}}^{-;p_1+1,p_2}\Big\}. \tag{169}$$

Every seed here has arguments (u,u), and its superscript $\pm$ means $\kappa = \pm 1/2$.

Differentiating the seed definition and integrating the resulting time derivatives by parts gives

$$\frac{\mathrm{d}}{\mathrm{d}r}\mathcal{K}_{\mathsf{a}\mathsf{b}}^{\kappa;p_1,p_2}(u(r),u(r))\Big|_{r=1} = \mathsf{i}\mathsf{a}\,\mathcal{K}_{\mathsf{a}\mathsf{b}}^{\kappa;p_1+1,p_2}(1,1) + \mathsf{i}\mathsf{b}\,\mathcal{K}_{\mathsf{a}\mathsf{b}}^{\kappa;p_1,p_2+1}(1,1). \tag{170}$$

Both the explicit powers of r and the factors $(\mathsf{a}\pm r)/r$ in (168) must also be differentiated.

Then the coefficients used in the main text are

$$\begin{aligned}\mathcal{P}_{t,1} &= 2^{2\mathrm{i}\tilde{\nu}}\operatorname*{FP}_{\delta=0}\left[\mathcal{H}_0(\mathrm{i}\tilde{\nu}+\delta,\mathrm{i}\tilde{\nu}+\delta;1) + \frac{2\mathrm{i}}{1+2\mathrm{i}\tilde{\nu}}\mathcal{H}_1(1+\mathrm{i}\tilde{\nu}+\delta,\mathrm{i}\tilde{\nu}+\delta;1)\right],\\ \mathcal{P}_{t,2} &= 2^{2\mathrm{i}\tilde{\nu}}\operatorname*{FP}_{\delta=0}(\partial_r-1)\mathcal{H}_0(\mathrm{i}\tilde{\nu}+\delta,\mathrm{i}\tilde{\nu}+\delta;r)|_{r=1},\end{aligned} \tag{171}$$

The factor $2^{2\mathrm{i}\tilde{\nu}}$ converts the seed normalization at k_L to the $k_{12} \simeq 2k_L$ convention in (144).

Here $\mathrm{FP}_{\delta=0}$ takes the constant Laurent coefficient after summing all SK and $\kappa = \pm 1/2$ branches with the same regulator δ in both time exponents.

After combining the SK branches, the finite coefficients can be expressed using a single ordinary hypergeometric function,

$$\begin{aligned}\mathcal{F}_t(\tilde{\nu}) &= {}_3F_2\left[\begin{matrix}1,\ 1+\mathrm{i}\tilde{\nu},\ 1+2\mathrm{i}\tilde{\nu}\\ 2+2\mathrm{i}\tilde{\nu},\ 2+2\mathrm{i}\tilde{\nu}\end{matrix}\middle|1\right],\\ \mathcal{E}_t(\tilde{\nu}) &= \frac{\pi^2\Gamma(1+2\mathrm{i}\tilde{\nu})}{\cosh(\pi\tilde{\nu})\Gamma^2(1+\mathrm{i}\tilde{\nu})}.\end{aligned} \tag{172}$$

The resulting expressions are

$$\begin{aligned}\mathcal{P}_{t,1} &= \Gamma(1+2\mathrm{i}\tilde{\nu})\cosh(\pi\tilde{\nu})\left[\mathcal{E}_t - \frac{2\mathcal{F}_t}{(1+2\mathrm{i}\tilde{\nu})^2} - \frac{(5+2\mathrm{i}\tilde{\nu})(4+9\mathrm{i}\tilde{\nu}-4\tilde{\nu}^2)}{2(1+\mathrm{i}\tilde{\nu})(2+\mathrm{i}\tilde{\nu})(1+2\mathrm{i}\tilde{\nu})}\right],\\ \mathcal{P}_{t,2} &= \Gamma(1+2\mathrm{i}\tilde{\nu})\cosh(\pi\tilde{\nu})\left[\frac{2\mathcal{F}_t}{1+2\mathrm{i}\tilde{\nu}} - (1+2\mathrm{i}\tilde{\nu})\mathcal{E}_t + \frac{\mathrm{i}\tilde{\nu}(5+2\mathrm{i}\tilde{\nu})}{2(1+\mathrm{i}\tilde{\nu})(2+\mathrm{i}\tilde{\nu})}\right].\end{aligned} \tag{173}$$

A.3 Angular Coefficients

Using $\mathcal{G}_t$ defined in (161), define the common soft factor

$$\mathcal{B}_t(\tilde{\nu}) = \frac{\Gamma(\frac{1}{2}-\mathrm{i}\tilde{\nu})^4\mathcal{G}_t(\tilde{\nu})}{(4\pi)^{7/2}(5+2\mathrm{i}\tilde{\nu})(3+2\mathrm{i}\tilde{\nu})(5+4\mathrm{i}\tilde{\nu})(3+4\mathrm{i}\tilde{\nu})}. \tag{174}$$

Substituting (162) gives

$$\begin{aligned}
A_{1,t} &= -2\mathcal{B}_t(1+\mathrm{i}\tilde{\nu})(1+2\mathrm{i}\tilde{\nu}) \\
&\quad\times\left[(5+2\mathrm{i}\tilde{\nu})(3\mathcal{P}_{t,1}^2+2\mathcal{P}_{t,1}\mathcal{P}_{t,2})+(2+\mathrm{i}\tilde{\nu})\mathcal{P}_{t,2}^2\right], \\
A_{2,t} &= 2\mathcal{B}_t\mathcal{P}_{t,2}^2(1+\mathrm{i}\tilde{\nu})^2(1+2\mathrm{i}\tilde{\nu}), \qquad A_{3,t} = -3A_{2,t}, \\
B_t &= \mathcal{B}_t\mathcal{P}_{t,2}^2(1+\mathrm{i}\tilde{\nu})(1+2\mathrm{i}\tilde{\nu})(3+2\mathrm{i}\tilde{\nu}), \\
D_t &= -2\mathcal{B}_t\mathcal{P}_{t,2}^2(1+\mathrm{i}\tilde{\nu})(2+\mathrm{i}\tilde{\nu})(3+2\mathrm{i}\tilde{\nu}).
\end{aligned} \tag{175}$$

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